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4 Revision History

Changes from Original (June 2018) to Revision A

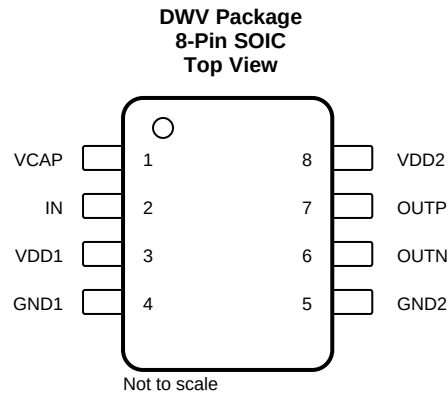
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• Changed document status from Advance Information to Production Data	1
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5 Device Comparison Table

PARAMETER	ISO224B	ISO224A
Input offset voltage, V_{OS}	± 5 mV (max)	± 50 mV (max)
Input offset drift, TCV_{OS}	± 15 $\mu V/^{\circ}C$ (max)	± 60 $\mu V/^{\circ}C$ (max)
Input-referred noise	3 $\mu V/\sqrt{Hz}$ (typ)	4 $\mu V/\sqrt{Hz}$ (typ)
Gain error, E_G	$\pm 0.3\%$ (max)	$\pm 1\%$ (max)
Gain error drift, TCE_G	± 35 ppm/ $^{\circ}C$ (max)	± 60 ppm/ $^{\circ}C$ (max)
Nonlinearity	$\pm 0.01\%$ (max)	$\pm 0.02\%$ (max)
Output bandwidth, BW	275 kHz (typ)	185 kHz (typ)
Common-mode transient immunity, CMTI	80 kV/ μs (typ)	30 kV/ μs (typ)
IN to OUTP, OUTN signal delay (50% – 50%)	2.2 μs (typ)	2.8 μs (typ)

6 Pin Configuration and Functions



Pin Functions

PIN		I/O	DESCRIPTION
NO.	NAME		
1	VCAP	—	Supply decoupling capacitor. Connect a 0.22- μF capacitor between this pin and the high-side analog ground.
2	IN	I	Analog input
3	VDD1	—	High-side power supply, 4.5 V to 18 V. See the Power Supply Recommendations section for decoupling recommendations.
4	GND1	—	High-side analog ground
5	GND2	—	Low-side analog ground
6	OUTN	O	Inverting analog output
7	OUTP	O	Noninverting analog output
8	VDD2	—	Low-side power supply, 4.5 V to 5.5 V. See the Power Supply Recommendations section for decoupling recommendations.

7 Specifications

7.1 Absolute Maximum Ratings

see ⁽¹⁾

		MIN	MAX	UNIT
Power-supply voltage	VDD1 to GND1	–0.3	26	V
	VDD2 to GND2	–0.3	6.5	
Input voltage	IN to GND1 ⁽²⁾	–15	15	V
Input current	Continuous, at IN pin ⁽³⁾	–10	10	mA
Output voltage	OUTP, OUTN	GND2 – 0.3	VDD2 + 0.3	V
Temperature	Junction, T _J		150	°C
	Storage, T _{stg}	–65	150	

- (1) Stresses beyond those listed under *Absolute Maximum Ratings* may cause permanent damage to the device. These are stress ratings only, which do not imply functional operation of the device at these or any other conditions beyond those indicated under *Recommended Operating Conditions*. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.
- (2) Exposure to absolute-maximum-rated condition for extended periods may increase input leakage current.
- (3) Limit the input current at IN pin to prevent permanent damage to the device. The IN pin is internally protected by a voltage clamp. See [Figure 42](#) for a typical current versus voltage characteristic curve of the input clamp.

7.2 ESD Ratings

				VALUE	UNIT
V _(ESD)	Electrostatic discharge	Human-body model (HBM), per ANSI/ESDA/JEDEC JS-001 ⁽¹⁾	IN pin only	±9000	V
			All pins except IN	±3000	
		Charged-device model (CDM), per JEDEC specification JESD22-C101 ⁽²⁾		±1000	

- (1) JEDEC document JEP155 states that 500-V HBM allows safe manufacturing with a standard ESD control process.
- (2) JEDEC document JEP157 states that 250-V CDM allows safe manufacturing with a standard ESD control process.

7.3 Recommended Operating Conditions

over operating ambient temperature range (unless otherwise noted)

			MIN	NOM	MAX	UNIT
POWER SUPPLY						
	High-side power supply	VDD1 to GND1	4.5	5	18	V
	Low-side power supply	VDD2 to GND2	4.5	5	5.5	V
ANALOG INPUT						
V _{Clipping}	Input voltage before clipping output ⁽¹⁾	IN to GND1		±13.8		V
V _{FSR}	Specified linear input full-scale voltage ⁽¹⁾	IN to GND1	–12		12	V
TEMPERATURE RANGE						
T _A	Specified ambient temperature		–55	25	125	°C

- (1) See the [Analog Input](#) section for more details.

7.4 Thermal Information

THERMAL METRIC ⁽¹⁾		ISO224x	UNIT
		DWV (SOIC)	
		8 PINS	
$R_{\theta JA}$	Junction-to-ambient thermal resistance	96.3	°C/W
$R_{\theta JC(top)}$	Junction-to-case (top) thermal resistance	36.9	°C/W
$R_{\theta JB}$	Junction-to-board thermal resistance	60.1	°C/W
Ψ_{JT}	Junction-to-top characterization parameter	16.9	°C/W
Ψ_{JB}	Junction-to-board characterization parameter	58.2	°C/W
$R_{\theta JC(bot)}$	Junction-to-case (bottom) thermal resistance	n/a	°C/W

(1) For more information about traditional and new thermal metrics, see the [Semiconductor and IC Package Thermal Metrics](#) application report.

7.5 Power Ratings⁽¹⁾

PARAMETER		TEST CONDITIONS	VALUE	UNIT
P_D	Maximum power dissipation (both sides)	VDD1 = 18 V, VDD2 = 5.5 V	194.9	mW
		VDD1 = VDD2 = 5.5 V	97.4	
P_{D1}	Maximum power dissipation (high-side supply)	VDD1 = 18 V	140.4	mW
		VDD1 = 5.5 V	42.9	
P_{D2}	Maximum power dissipation (low-side supply)	VDD2 = 5.5 V	54.5	mW

(1) See the [Electrical Characteristics](#) table for maximum supply current specifications.

7.6 Insulation Specifications

over operating ambient temperature range (unless otherwise noted)

PARAMETER		TEST CONDITIONS	VALUE	UNIT
GENERAL				
CLR	External clearance ⁽¹⁾	Shortest pin-to-pin distance through air	≥ 8.5	mm
CPG	External creepage ⁽¹⁾	Shortest pin-to-pin distance across the package surface	≥ 8.5	mm
DTI	Distance through insulation	Minimum internal gap (internal clearance) of the double insulation	≥ 0.021	mm
CTI	Comparative tracking index	DIN EN 60112 (VDE 0303-11); IEC 60112	≥ 600	V
	Material group	According to IEC 60664-1	I	
	Overvoltage category per IEC 60664-1	Rated mains voltage ≤ 600 V _{RMS}	I-IV	
		Rated mains voltage ≤ 1000 V _{RMS}	I-III	
DIN VDE V 0884-11 (VDE V 0884-11): 2017-01 ⁽²⁾				
V _{IORM}	Maximum repetitive peak isolation voltage	At AC voltage (bipolar or unipolar)	2121	V _{PK}
V _{IOWM}	Maximum-rated isolation working voltage	At AC voltage (sine wave); time dependent dielectric breakdown (TDDB) test; see Figure 4	1500	V _{RMS}
		At DC voltage	2121	V _{DC}
V _{IOTM}	Maximum transient isolation voltage	V _{TEST} = V _{IOTM} , t = 60 s (qualification test)	7071	V _{PK}
		V _{TEST} = 1.2 × V _{IOTM} , t = 1 s (100% production test)	8485	
V _{IOSM}	Maximum surge isolation voltage ⁽³⁾	Test method per IEC 60065, 1.2/50-μs waveform, V _{TEST} = 1.6 × V _{IOSM} = 12800 V _{PK} (qualification)	8000	V _{PK}
q _{pd}	Apparent charge ⁽⁴⁾	Method A, after input/output safety test subgroup 2/3, V _{ini} = V _{IOTM} , t _{ini} = 60 s, V _{pd(m)} = 1.2 × V _{IORM} = 2545 V _{PK} , t _m = 10 s	≤ 5	pC
		Method A, after environmental tests subgroup 1, V _{ini} = V _{IOTM} , t _{ini} = 60 s, V _{pd(m)} = 1.6 × V _{IORM} = 3394 V _{PK} , t _m = 10 s	≤ 5	
		Method B1, at routine test (100% production) and preconditioning (type test), V _{ini} = V _{IOTM} , t _{ini} = 1 s, V _{pd(m)} = 1.875 × V _{IORM} = 3977 V _{PK} , t _m = 1 s	≤ 5	
C _{IO}	Barrier capacitance, input to output ⁽⁵⁾	V _{IO} = 0.5 V _{PP} at 1 MHz	~1	pF
R _{IO}	Insulation resistance, input to output ⁽⁵⁾	V _{IO} = 500 V at T _A = 25°C	> 10 ¹²	Ω
		V _{IO} = 500 V at 100°C ≤ T _A ≤ 125°C	> 10 ¹¹	
		V _{IO} = 500 V at T _S = 150°C	> 10 ⁹	
	Pollution degree		2	
	Climatic category		55/125/21	
UL1577				
V _{ISO}	Withstand isolation voltage	V _{TEST} = V _{ISO} = 5000 V _{RMS} or 7071 V _{DC} , t = 60 s (qualification), V _{TEST} = 1.2 × V _{ISO} = 6000 V _{RMS} , t = 1 s (100% production test)	5000	V _{RMS}

- (1) Apply creepage and clearance requirements according to the specific equipment isolation standards of an application. Care must be taken to maintain the creepage and clearance distance of a board design to ensure that the mounting pads of the isolator on the printed circuit board (PCB) do not reduce this distance. Creepage and clearance on a PCB become equal in certain cases. Techniques such as inserting grooves and ribs on the PCB are used to help increase these specifications.
- (2) This coupler is suitable for *safe electrical insulation* only within the safety ratings. Compliance with the safety ratings shall be ensured by means of suitable protective circuits.
- (3) Testing is carried out in air or oil to determine the intrinsic surge immunity of the isolation barrier.
- (4) Apparent charge is electrical discharge caused by a partial discharge (pd).
- (5) All pins on each side of the barrier are tied together, creating a two-pin device.

7.7 Safety-Related Certifications

VDE	UL
Certified according to DIN VDE V 0884-11 (VDE V 0884-11): 2017-01, DIN EN 60950-1 (VDE 0805 Teil 1): 2014-08, and DIN EN 60065 (VDE 0860): 2005-11	Recognized under 1577 component recognition and CSA component acceptance NO 5 programs
Reinforced insulation	Single protection
File number: 40040142	File number: E181974

7.8 Safety Limiting Values

Safety limiting intends to minimize potential damage to the isolation barrier upon failure of input or output circuitry.

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
I _S	Safety input, output, or supply current	R _{θJA} = 96.3°C/W, T _J = 150°C, T _A = 25°C, VDD1 = 18 V, VDD2 = 5.5 V, see Figure 2			55	mA
		R _{θJA} = 96.3°C/W, T _J = 150°C, T _A = 25°C, VDD1 = VDD2 = 5.5 V, see Figure 2			236	
P _S	Safety input, output, or total power	R _{θJA} = 96.3°C/W, T _J = 150°C, T _A = 25°C, see Figure 3			1298 ⁽¹⁾	mW
T _S	Maximum safety temperature				150	°C

- (1) The maximum safety temperature, T_S , has the same value as the maximum junction temperature, T_J , specified for the device. The I_S and P_S parameters represent the safety current and safety power, respectively. Do not exceed the maximum limits of I_S and P_S . These limits vary with the ambient temperature, T_A .

The junction-to-air thermal resistance, $R_{\theta JA}$, in the [Thermal Information](#) table is that of a device installed on a high-K test board for leaded surface-mount packages. Use these equations to calculate the value for each parameter:

$T_J = T_A + R_{\theta JA} \times P$, where P is the power dissipated in the device.

$T_{J(\text{max})} = T_S = T_A + R_{\theta JA} \times P_S$, where $T_{J(\text{max})}$ is the maximum junction temperature.

$P_S = I_S \times V_{DD1_{\text{max}}} + I_S \times V_{DD2_{\text{max}}}$, where $V_{DD1_{\text{max}}}$ is the maximum high-side supply voltage and $V_{DD2_{\text{max}}}$ is the maximum low-side supply voltage.

7.9 Electrical Characteristics

minimum and maximum specifications apply from $T_A = -55^{\circ}\text{C}$ to $+125^{\circ}\text{C}$, $V_{DD1} = 4.5\text{ V}$ to 18 V , $V_{DD2} = 4.5\text{ V}$ to 5.5 V , $V_{IN} = -12\text{ V}$ to 12 V , and $R_{LOAD} = 10\text{ k}\Omega$; typical specifications are at $T_A = 25^{\circ}\text{C}$, and $V_{DD1} = V_{DD2} = 5\text{ V}$ (unless otherwise noted)

PARAMETER	TEST CONDITIONS	MIN	TYP	MAX	UNIT
ANALOG INPUT					
V_{OS}	Input offset voltage ⁽¹⁾	Initial, at $T_A = 25^{\circ}\text{C}$, $IN = \text{GND1}$, ISO224B	-5	± 1	5
		Initial, at $T_A = 25^{\circ}\text{C}$, $IN = \text{GND1}$, ISO224A	-50	± 1	50
TCV_{OS}	Input offset voltage drift ⁽¹⁾	ISO224B	-15	± 3	15
		ISO224A	-60	± 12	60
C_{IN}	Input capacitance	IN to GND1		2	pF
R_{IN}	Input resistance	IN to GND1	1	1.25	M Ω
I_{IB}	Input bias current	IN = GND1		± 15	nA
TCI_{IB}	Input bias current drift	IN = GND1		± 30	pA/ $^{\circ}\text{C}$
e_n	Input-referred noise density	ISO224B		3	$\mu\text{V}/\sqrt{\text{Hz}}$
		ISO224A		4	

- (1) The typical value includes one sigma statistical variation.

Electrical Characteristics (continued)

minimum and maximum specifications apply from $T_A = -55^{\circ}\text{C}$ to $+125^{\circ}\text{C}$, $V_{DD1} = 4.5\text{ V}$ to 18 V , $V_{DD2} = 4.5\text{ V}$ to 5.5 V , $V_{IN} = -12\text{ V}$ to 12 V , and $R_{LOAD} = 10\text{ k}\Omega$; typical specifications are at $T_A = 25^{\circ}\text{C}$, and $V_{DD1} = V_{DD2} = 5\text{ V}$ (unless otherwise noted)

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT	
ANALOG OUTPUTS							
	Nominal gain	(V _{OUTP} – V _{OUTN}) / V _{IN}	1/3			V/V	
E _G	Gain error ⁽¹⁾	Initial, at T _A = 25°C, ISO224B	–0.3%	±0.05%	0.3%		
		Initial, at T _A = 25°C, ISO224A	–1%	0.4%	1%		
TCE _G	Gain error drift ⁽¹⁾	ISO224B	–35	±10	35	ppm/°C	
		ISO224A	–60	±20	60		
	Nonlinearity	ISO224B	–0.01%	±0.003%	0.01%		
		ISO224A	–0.02%	±0.003%	0.02%		
	Nonlinearity drift		±0.1			ppm/°C	
THD	Total harmonic distortion	f _{IN} = 10 kHz	–84			dB	
	Output noise	IN = GND1, f _{IN} = 0 Hz, BW = 10 kHz	300			μV _{RMS}	
		IN = GND1, f _{IN} = 0 Hz, BW = 100 kHz	360				
PSRR	Power-supply rejection ratio ⁽²⁾	vs VDD1, at DC	–107			dB	
		vs VDD1, 100-mV and 10-kHz ripple	–101				
		vs VDD2, at DC	–71				
		vs VDD2, 100-mV and 10-kHz ripple	–56				
V _{OUT}	Output voltage	OUTP or OUTN to GND2	GND2 + 0.2	VDD2 – 0.2		V	
V _{CMout}	Common-mode output voltage	(V _{OUTP} + V _{OUTN}) / 2	0.48 × VDD2	VDD2 / 2	0.52 × VDD2	V	
V _{FAILSAFE}	Failsafe output voltage	VDD1 missing, OUTP and OUTN forced to GND2	GND2 + 0.1			V	
I _{SC}	Output short-circuit current	On OUTP or OUTN to GND2	±18			mA	
	Overload recovery time		5			μs	
R _{OUT}	Output resistance	On OUTP or OUTN to GND2	< 0.5			Ω	
C _{LOAD}	Capacitive load drive ⁽³⁾	On OUTP or OUTN to GND2	100			pF	
		OUTP to OUTN	50				
R _{LOAD}	Resistive load	On OUTP or OUTN	10			kΩ	
BW	Small signal output bandwidth	ISO224B	220	275			kHz
		ISO224A	150	185			
CMTI	Common-mode transient immunity	GND1 – GND2 = 1 kV, ISO224B	55	80			kV/μs
		GND1 – GND2 = 1 kV, ISO224A	15	30			
POWER SUPPLY							
I _{DD1}	High-side supply current		6.1			7.8	mA
I _{DD2}	Low-side supply current		7.8			9.9	mA

(2) This parameter is output referred.

(3) Use series resistor to decouple higher capacitive load.

7.10 Switching Characteristics

over operating ambient temperature range (unless otherwise noted)

PARAMETER	TEST CONDITIONS	MIN	TYP	MAX	UNIT
t_r, t_f	ISO224B on OUTP, OUTN		1.5		μs
	ISO224A on OUTP, OUTN		2		μs
IN to OUTP, OUTN signal delay (50% – 10%)	ISO224B, unfiltered output, see Figure 1		1.5	2	μs
	ISO224A, unfiltered output, see Figure 1		1.9	2.9	
IN to OUTP, OUTN signal delay (50% – 50%)	ISO224B, unfiltered output, see Figure 1		2.2	2.7	μs
	ISO224A, unfiltered output, see Figure 1		2.8	3.8	
IN to OUTP, OUTN signal delay (50% – 90%)	ISO224B, unfiltered output, see Figure 1		3	3.5	μs
	ISO224A, unfiltered output, see Figure 1		3.8	4.8	
t_{AS}	Analog startup time		250		μs

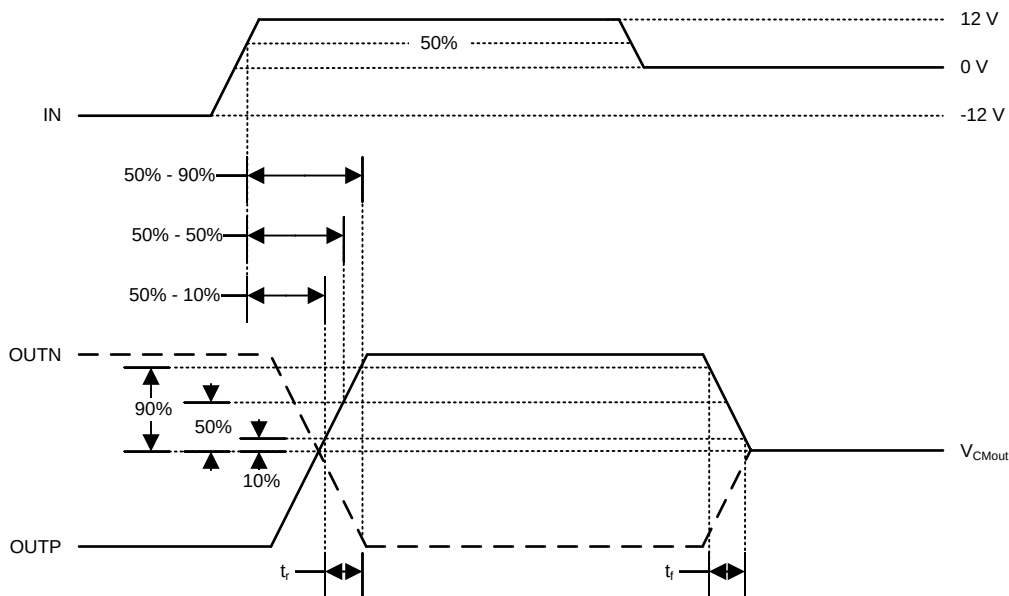


Figure 1. Delay Time Test Waveforms

7.11 Insulation Characteristics Curves

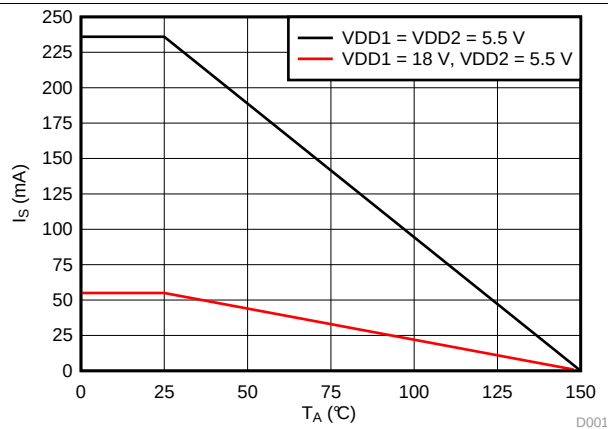


Figure 2. Thermal Derating Curve for Safety-Limiting Current per VDE

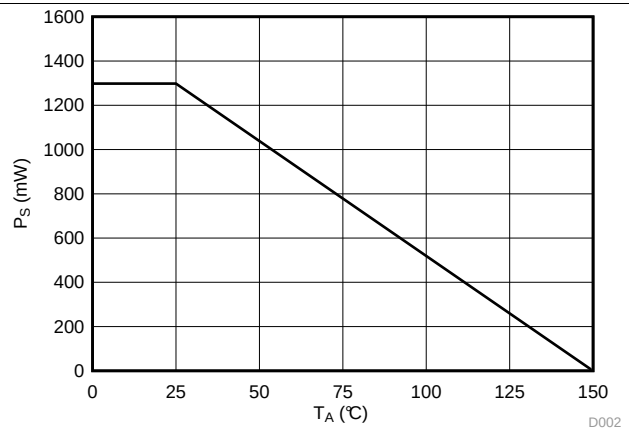


Figure 3. Thermal Derating Curve for Safety-Limiting Power per VDE

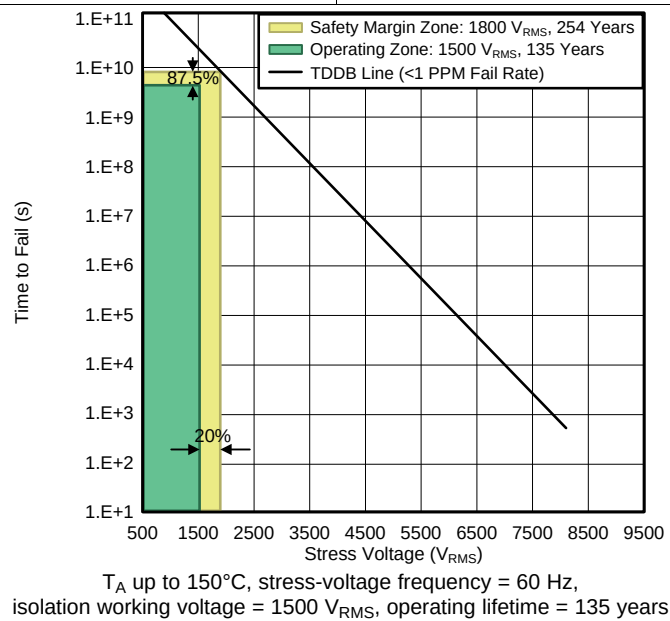


Figure 4. Reinforced Isolation Capacitor Lifetime Projection

7.12 Typical Characteristics

at $T_A = 25^\circ\text{C}$, $V_{DD1} = V_{DD2} = 5\text{ V}$, and $V_{INP} = -12\text{ V}$ to 12 V , unless otherwise noted.

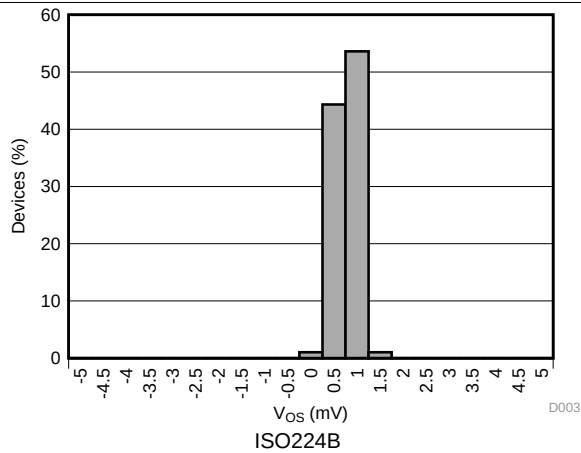


Figure 5. Input Offset Voltage Histogram

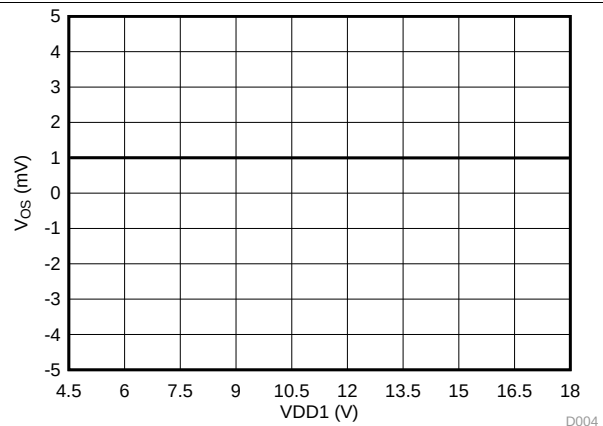


Figure 6. Input Offset Voltage vs High-Side Supply Voltage

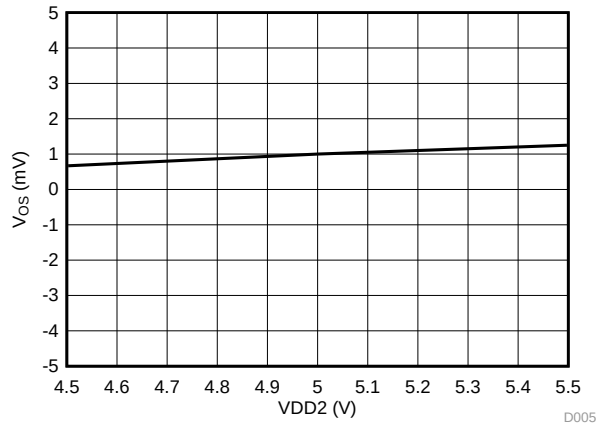


Figure 7. Input Offset Voltage vs Low-Side Supply Voltage

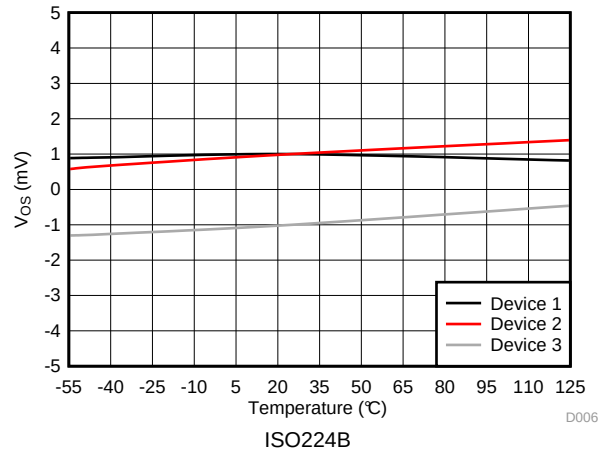


Figure 8. Input Offset Voltage vs Temperature

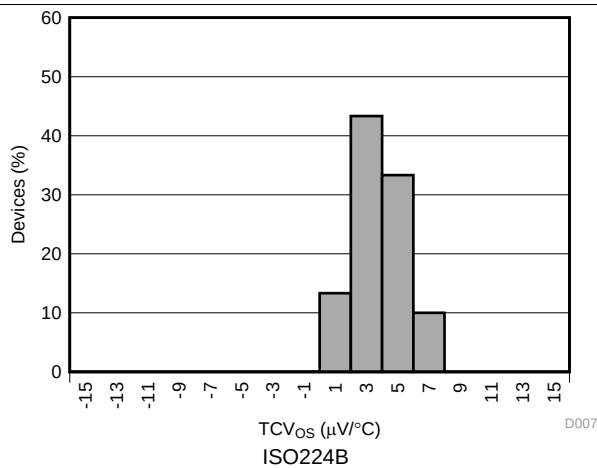


Figure 9. Input Offset Drift Histogram

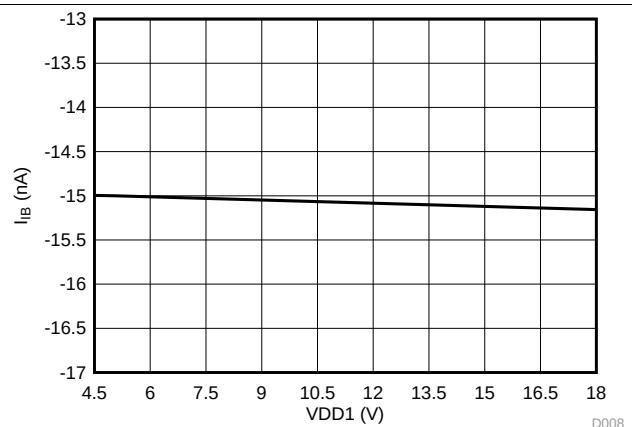


Figure 10. Input Bias Current vs High-Side Supply Voltage

Typical Characteristics (continued)

at $T_A = 25^\circ\text{C}$, $V_{DD1} = V_{DD2} = 5\text{ V}$, and $V_{INP} = -12\text{ V}$ to 12 V , unless otherwise noted.

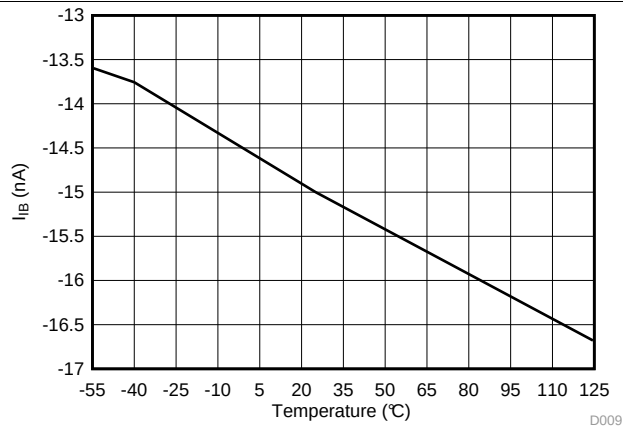


Figure 11. Input Bias Current vs Temperature

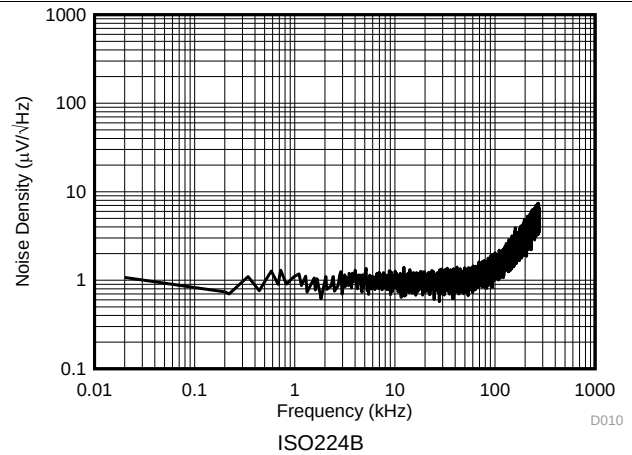


Figure 12. Input-Referred Noise Density vs Frequency

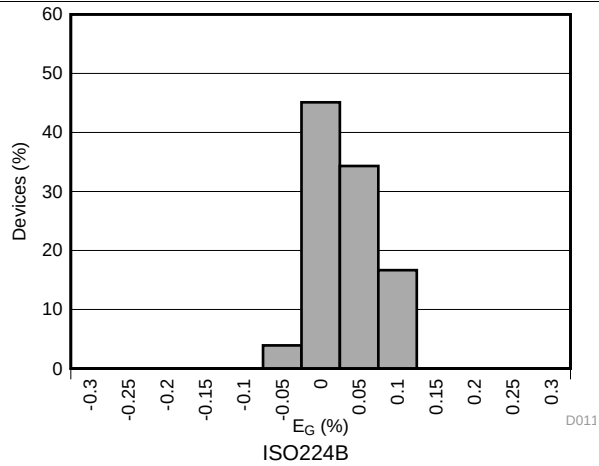


Figure 13. Gain Error Histogram

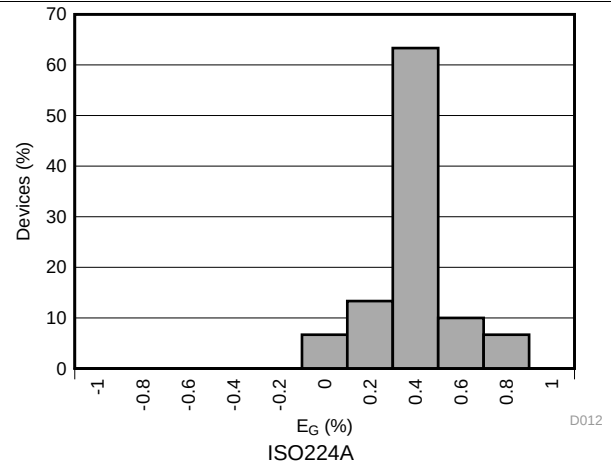


Figure 14. Gain Error Histogram

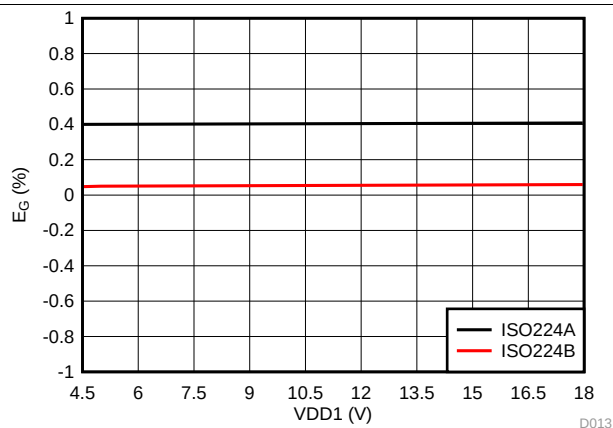


Figure 15. Gain Error vs High-Side Supply Voltage

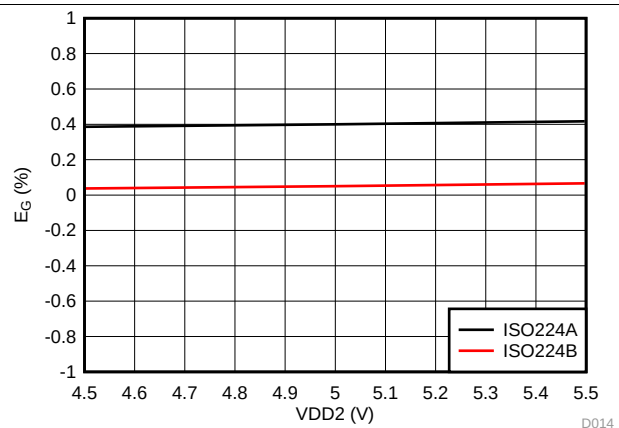


Figure 16. Gain Error vs Low-Side Supply Voltage

Typical Characteristics (continued)

at $T_A = 25^\circ\text{C}$, $V_{DD1} = V_{DD2} = 5\text{ V}$, and $V_{INP} = -12\text{ V to } 12\text{ V}$, unless otherwise noted.

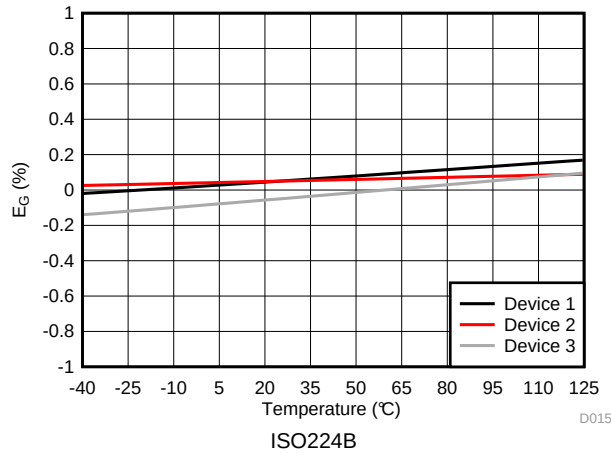


Figure 17. Gain Error vs Temperature

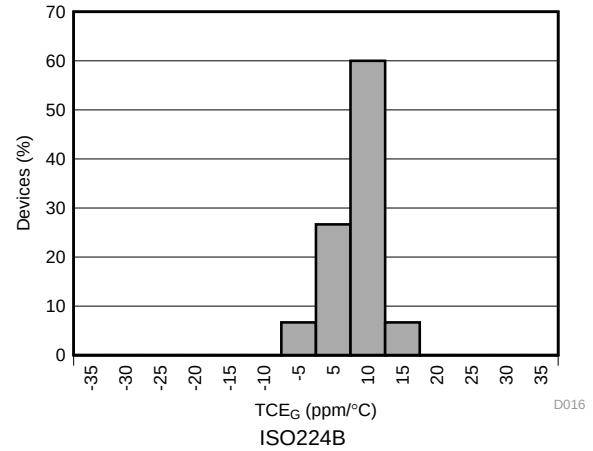


Figure 18. Gain Error Drift Histogram

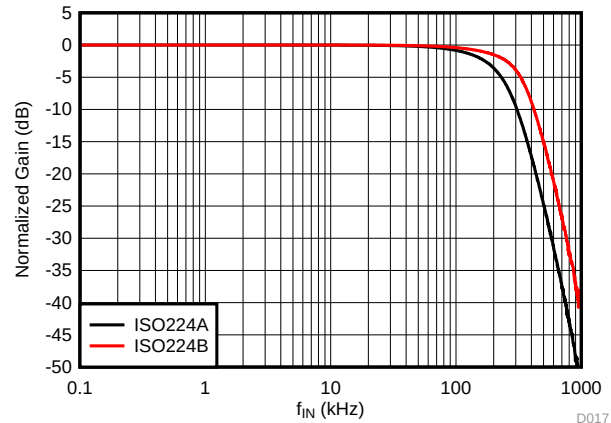


Figure 19. Normalized Gain vs Input Frequency

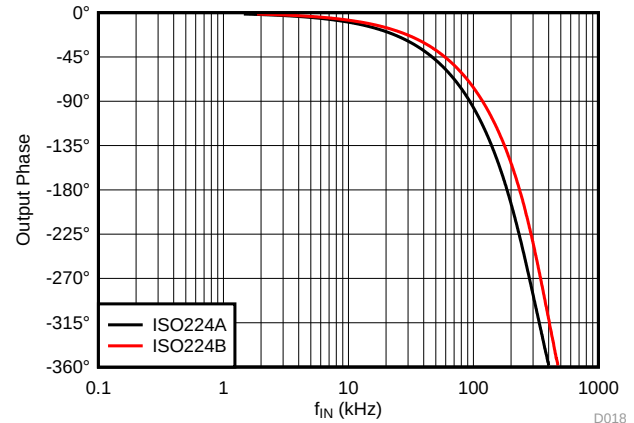


Figure 20. Output Phase vs Input Frequency

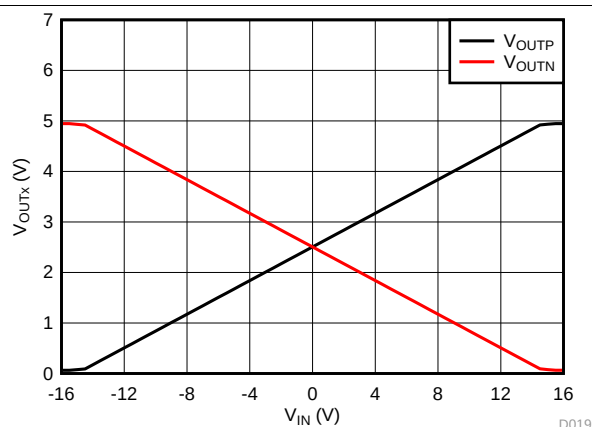


Figure 21. Output Voltage vs Input Voltage

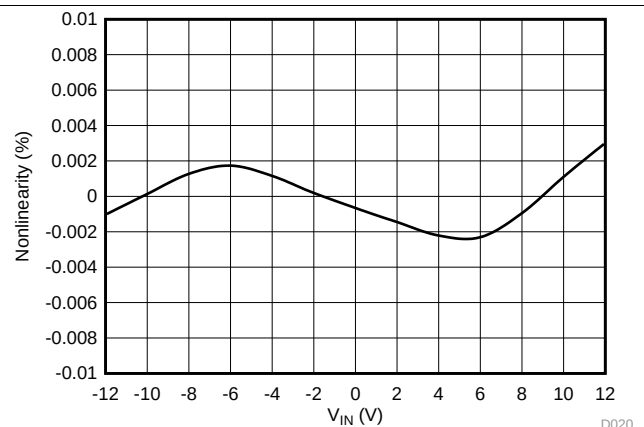


Figure 22. Nonlinearity vs Input Voltage

Typical Characteristics (continued)

at $T_A = 25^\circ\text{C}$, $V_{DD1} = V_{DD2} = 5\text{ V}$, and $V_{INP} = -12\text{ V}$ to 12 V , unless otherwise noted.

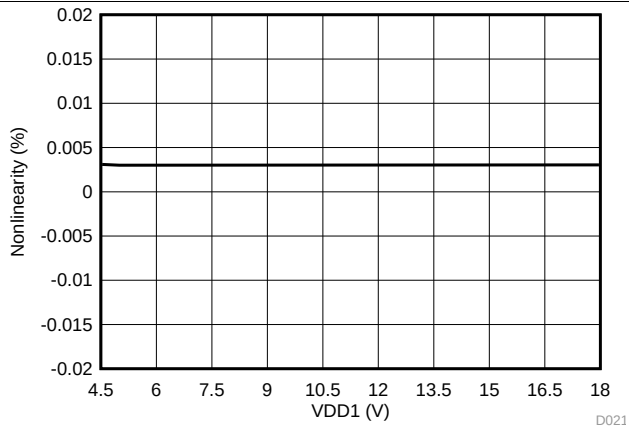


Figure 23. Nonlinearity vs High-Side Supply Voltage

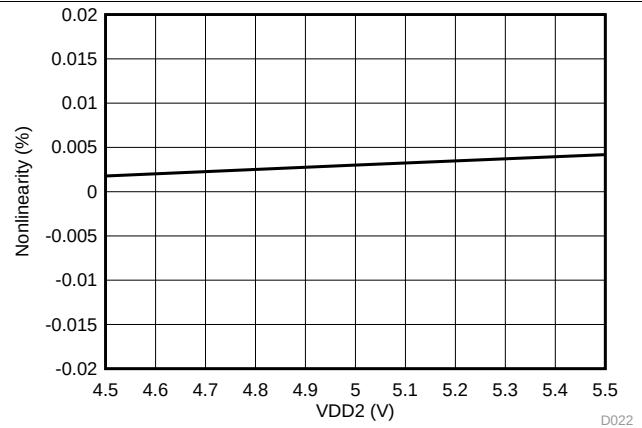


Figure 24. Nonlinearity vs Low-Side Supply Voltage

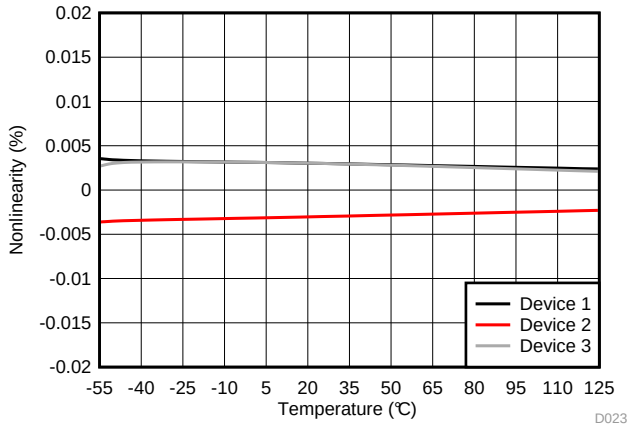


Figure 25. Nonlinearity vs Temperature

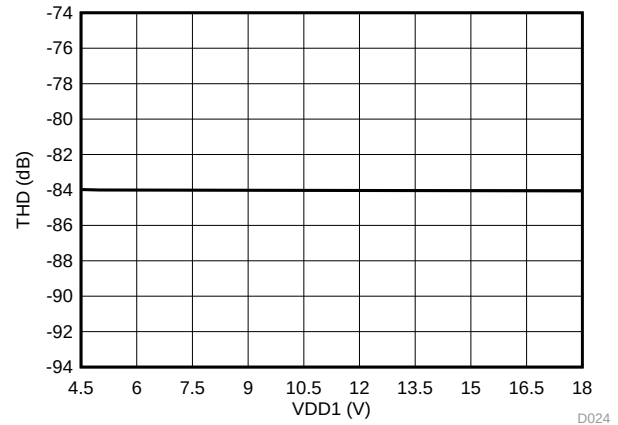


Figure 26. Total Harmonic Distortion vs High-Side Supply Voltage

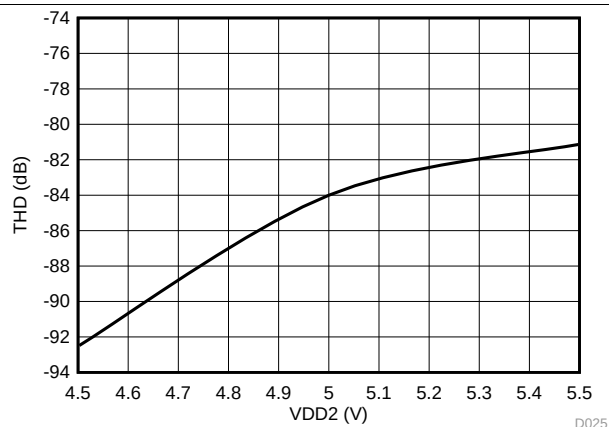


Figure 27. Total Harmonic Distortion vs Low-Side Supply Voltage

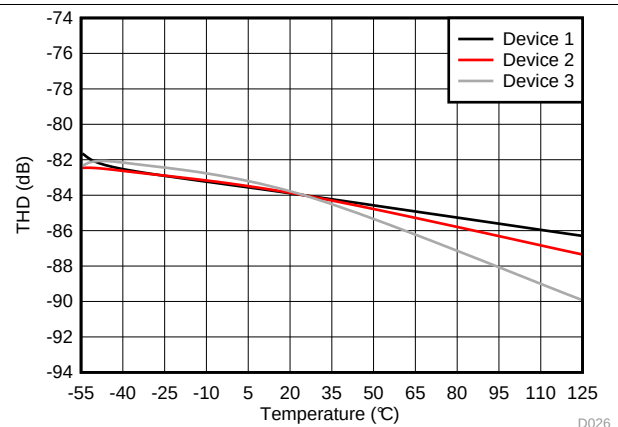


Figure 28. Total Harmonic Distortion vs Temperature

Typical Characteristics (continued)

at $T_A = 25^\circ\text{C}$, $V_{DD1} = V_{DD2} = 5\text{ V}$, and $V_{INP} = -12\text{ V}$ to 12 V , unless otherwise noted.

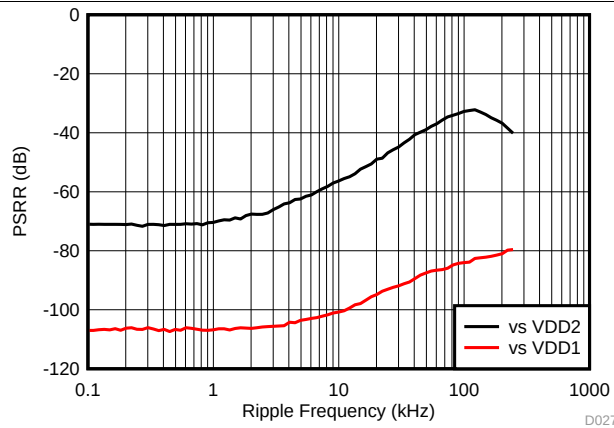


Figure 29. Power-Supply Rejection Ratio vs Ripple Frequency

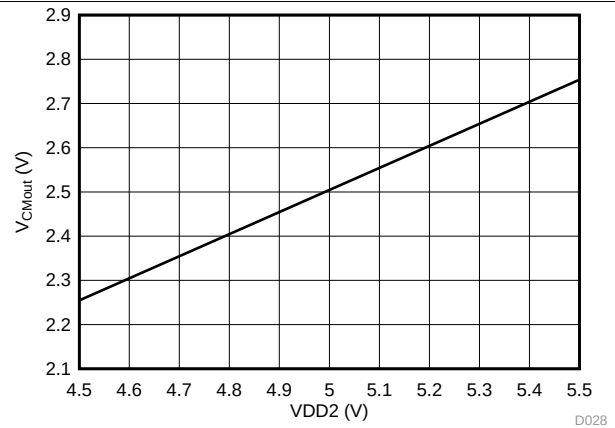


Figure 30. Output Common-Mode Voltage vs Low-Side Supply Voltage

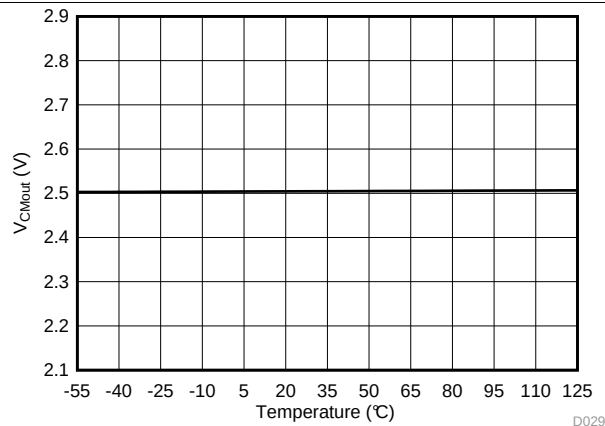


Figure 31. Output Common-Mode Voltage vs Temperature

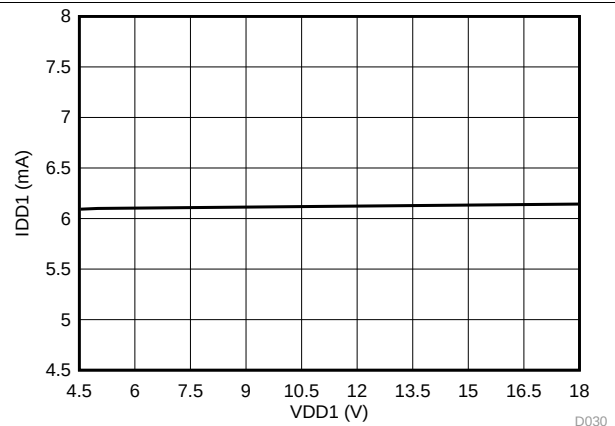


Figure 32. High-Side Supply Current vs High-Side Supply Voltage

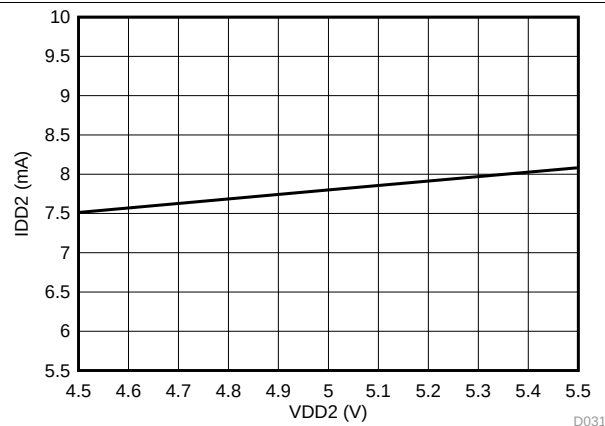


Figure 33. Low-Side Supply Current vs Low-Side Supply Voltage

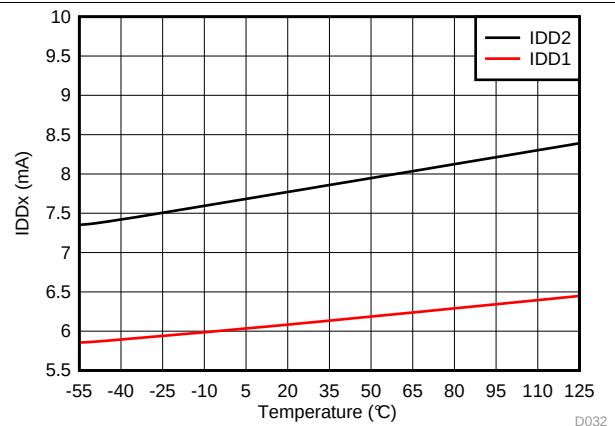


Figure 34. Supply Current vs Temperature

Typical Characteristics (continued)

at $T_A = 25^\circ\text{C}$, $V_{DD1} = V_{DD2} = 5\text{ V}$, and $V_{IN} = -12\text{ V}$ to 12 V , unless otherwise noted.

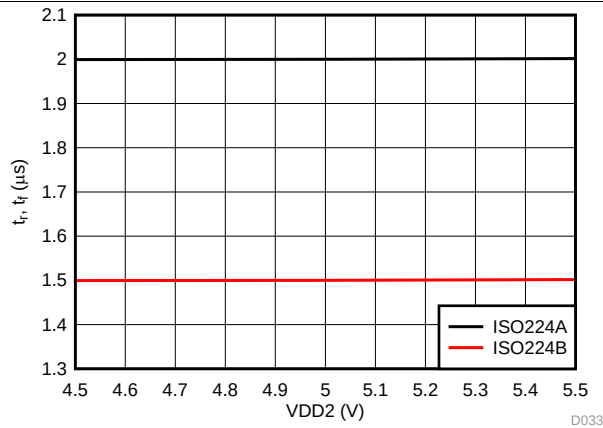


Figure 35. Output Rise and Fall Time vs Low-Side Supply Voltage

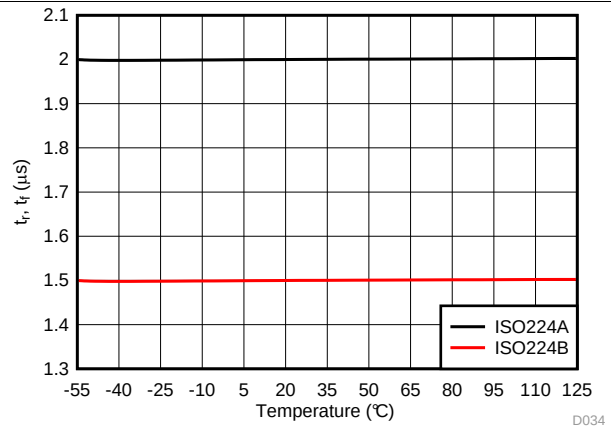


Figure 36. Output Rise and Fall Time vs Temperature

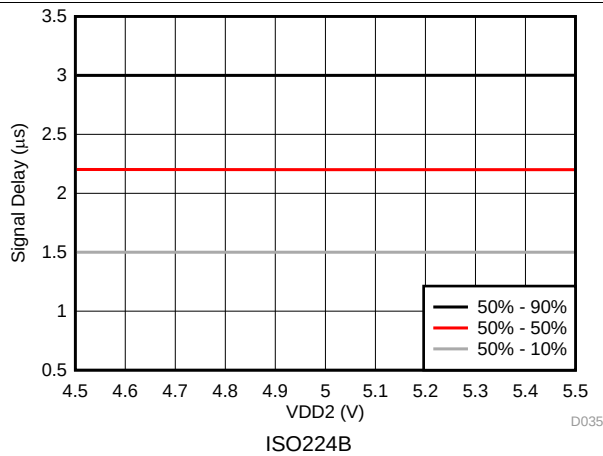


Figure 37. V_{IN} to V_{OUT} Signal Delay vs Low-Side Supply Voltage

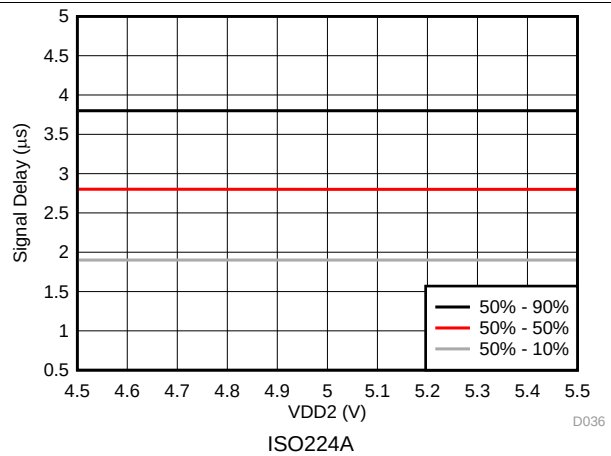


Figure 38. V_{IN} to V_{OUT} Signal Delay vs Low-Side Supply Voltage

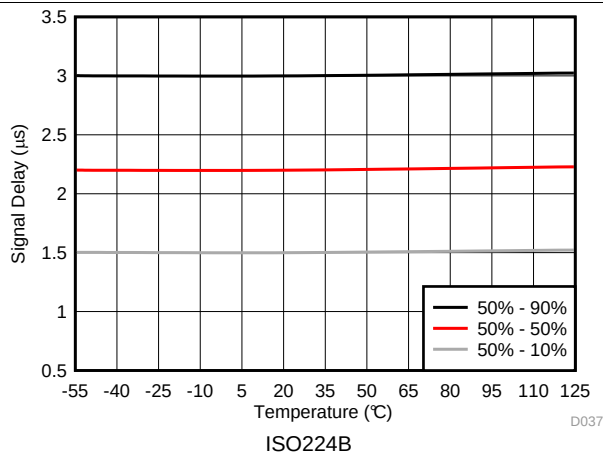


Figure 39. V_{IN} to V_{OUT} Signal Delay vs Temperature

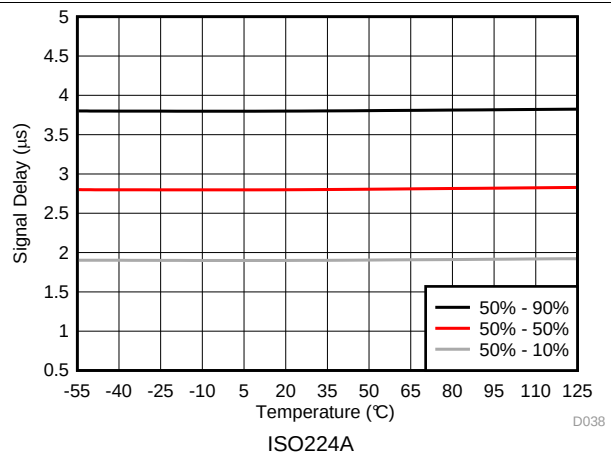


Figure 40. V_{IN} to V_{OUT} Signal Delay vs Temperature

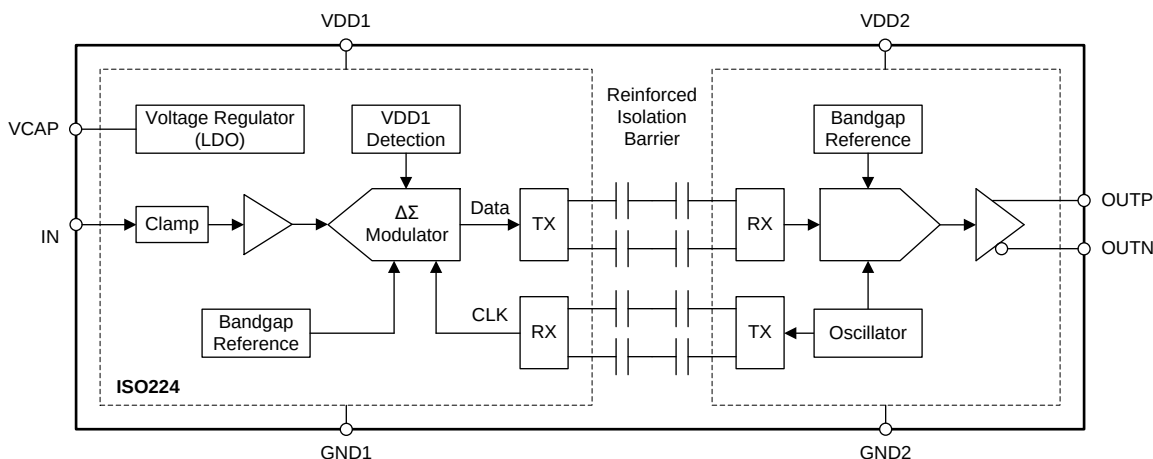
8 Detailed Description

8.1 Overview

The ISO224 is a precision, isolated amplifier with a high input impedance and wide input voltage range suitable for wide range of industrial applications. The input stage of the device drives a delta-sigma ($\Delta\Sigma$) modulator. The modulator uses the internal voltage reference and clock generator to convert the analog input signal to a digital bitstream. The drivers (called *TX* in the [Functional Block Diagram](#) section) transfer the output of the modulator across the isolation barrier that separates the high-side and low-side voltage domains. The received bitstream and clock are synchronized and processed by a digital-to-analog conversion stage on the low-side and presented as a differential analog output.

The SiO₂-based, double-capacitive isolation barrier supports a high level of magnetic field immunity, as described in [ISO72x Digital Isolator Magnetic-Field Immunity](#). The digital modulation used in the ISO224 and the isolation barrier characteristics result in high reliability and high common-mode transient immunity.

8.2 Functional Block Diagram



8.3 Feature Description

8.3.1 Analog Input

The input stage of the ISO224 feeds a switched capacitor, feed-forward $\Delta\Sigma$ modulator. The modulator converts the analog signal into a bitstream that is transferred over the isolation barrier, as described in the [Isolation Channel Signal Transmission](#) section. The high-impedance and low bias-current input of the ISO224 makes the device suitable for isolated voltage sensing applications.

Figure 41 visualizes the difference in the transfer function of the ISO224 depending on the input signal V_{IN} , as specified in the [Recommended Operating Conditions](#) table. With the input voltage within the specified full-scale range V_{FSR} , the output of the device changes in a linear way with small error as specified by the nonlinearity parameter in the [Electrical Characteristics](#) table. If the input signal exceeds the V_{FSR} range, the nonlinearity of the output signals increases and the amplitude clips at $V_{IN} = V_{CLIPPING}$.

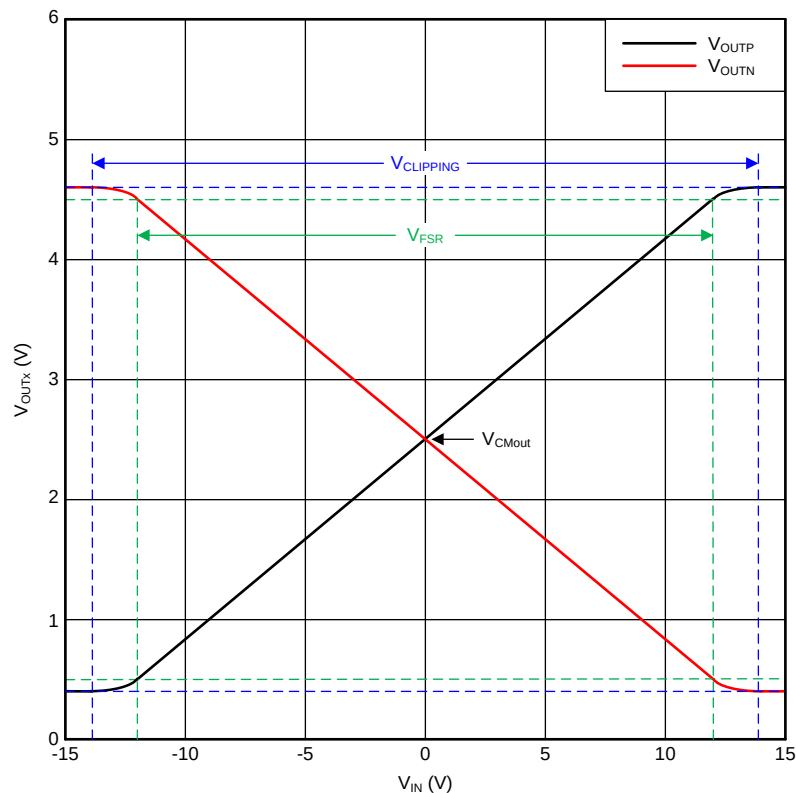


Figure 41. Transfer Function of the ISO224

There are two restrictions on the analog input signal at the IN pin. First, if the input voltage V_{IN} exceeds the range of -15 V to 15 V , the current must be limited to 10 mA to prevent damage to the input clamp, see the [Input Clamp Protection Circuit](#) section for further information. In addition, the linearity and noise performance of the ISO224 are ensured only when the analog input voltage remains within the specified linear full-scale range (V_{FSR}).

Feature Description (continued)

8.3.2 Input Clamp Protection Circuit

As illustrated in the [Functional Block Diagram](#), the ISO224 features an internal clamp protection circuit on the analog input IN. Using external protection circuits is recommended as a secondary protection scheme to protect the device against surges, ESD events, and electrical fast transient (EFT) conditions.

[Figure 42](#) shows a typical current versus voltage characteristic curve for the input clamp. Limit either the voltage V_{IN} at the input pin IN to the voltage range as defined in the [Recommended Operating Conditions](#) table or the input current to the limits as defined in the [Absolute Maximum Ratings](#) table.

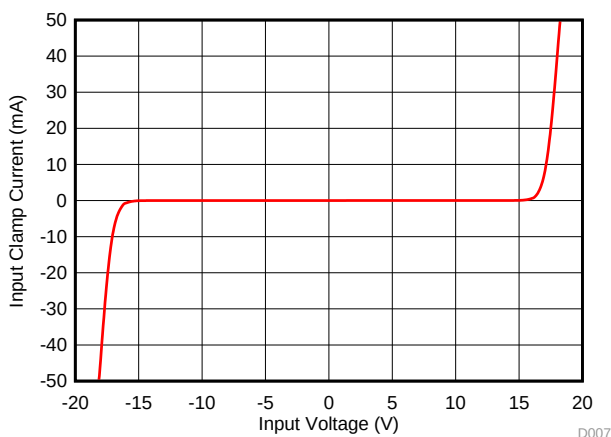


Figure 42. I-V Curve of the Input Clamp Protection Circuit

[Figure 43](#) shows a simple method to limit the input current with an external series resistor that is also used as part of the input low-pass filter.

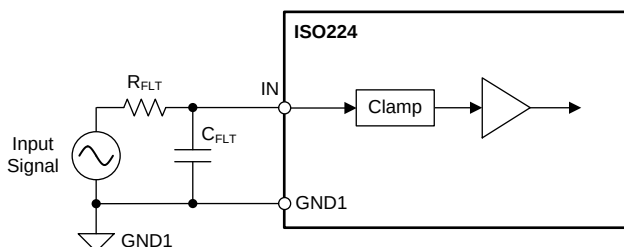


Figure 43. Series Resistor-Based Input Current Limitation on the Analog Inputs of ISO224

The input overvoltage protection clamp on the ISO224 is intended to control transient excursions on the input pins. Leaving the device in a state such that the clamp circuit is activated for extended periods of time in normal or power-down mode is not recommended because this fault condition can degrade device performance and reliability.

Feature Description (continued)

8.3.3 Isolation Channel Signal Transmission

The ISO224 uses an on-off keying (OOK) modulation scheme to transmit the modulator output bitstream across the SiO₂-based isolation barrier. As shown in [Figure 44](#), the transmitter modulates the bitstream at TX IN with an internally-generated, high-frequency carrier across the isolation barrier to represent a digital *one* and does not send a signal to represent the digital *zero*. The nominal frequency of the carrier used inside the ISO224 is 480 MHz.

The receiver demodulates the signal after advanced signal conditioning and produces the output. The ISO224 also incorporates advanced circuit techniques to maximize the CMTI performance and minimize the radiated emissions caused by the high-frequency carrier and IO buffer switching.

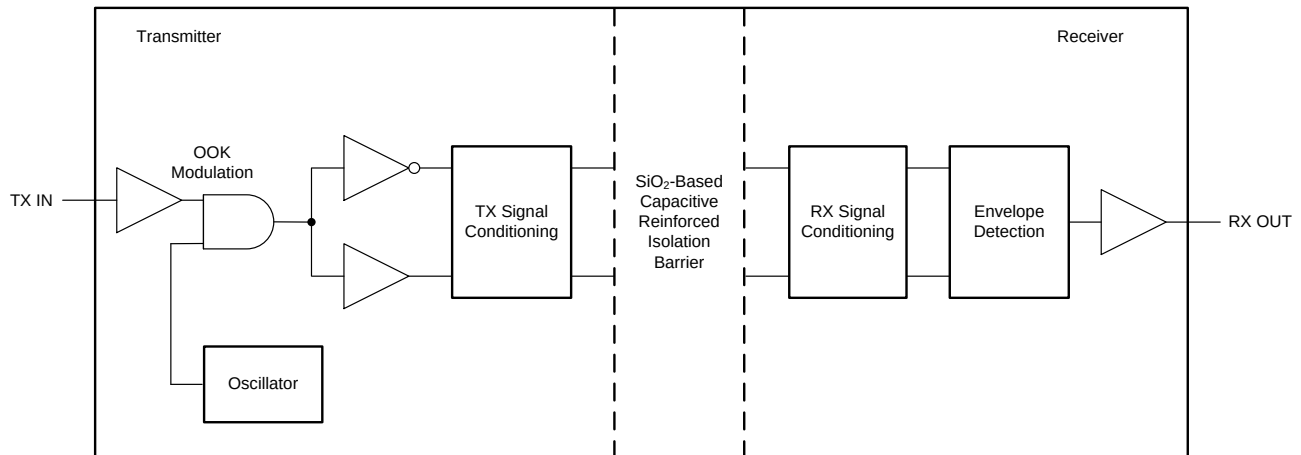


Figure 44. Block Diagram of an Isolation Channel

[Figure 45](#) shows the concept of the OOK scheme.

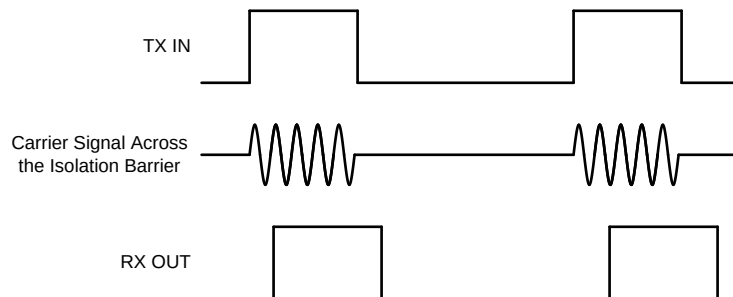


Figure 45. OOK-Based Modulation Scheme

Feature Description (continued)

8.3.4 Fail-Safe Output

The ISO224 offers a fail-safe output that simplifies diagnostics on system level. The fail-safe output is active when the high-side power supply VDD1 of the device is missing, independent of the input signal at the IN pin. Figure 46 shows that in that case both outputs, OUTP and OUTN, of the device are actively driven close to GND2 (see the V_{FAILSAFE} specification in the [Electrical Characteristics](#) table for details). For easy visualization, an example with the input signal $V_{\text{IN}} = 0 \text{ V}$ is shown for the valid VDD1 range of 4.5 V to 5.5 V.

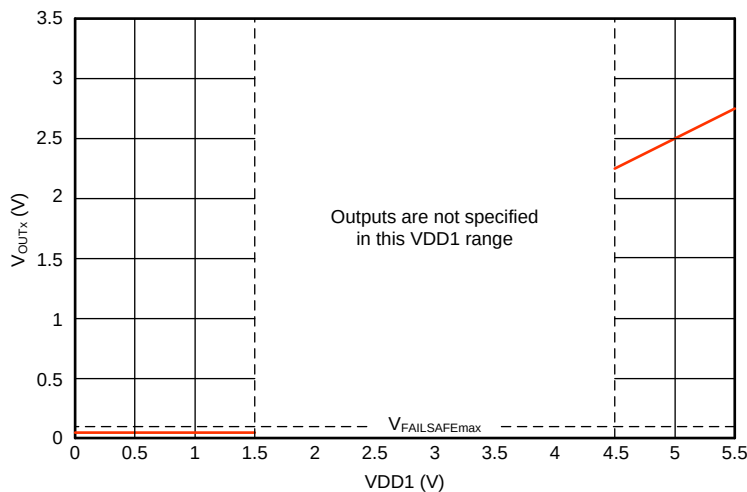


Figure 46. ISO224 Failsafe Output Behavior With $V_{\text{IN}} = 0 \text{ V}$

The [ISO224 Fail-Safe Output Feature application report](#) describes an example of a comparator-based circuit that detects the missing high-side supply in a system.

8.4 Device Functional Modes

The ISO224 is operational when the power supplies VDD1 and VDD2 are applied, as specified in the [Recommended Operating Conditions](#) table.

9 Application and Implementation

NOTE

Information in the following applications sections is not part of the TI component specification, and TI does not warrant its accuracy or completeness. TI's customers are responsible for determining suitability of components for their purposes. Customers should validate and test their design implementation to confirm system functionality.

9.1 Application Information

The ISO224 enables high-precision measurement of ± 10 -V signals that are used in a harsh environment in a wide range of industrial applications. The high input resistance of the device simplifies the connection of its input to different sensors or other signal sources. The very low nonlinearity, AC and DC errors, and temperature drift make the ISO224 a robust, high-performance, isolated amplifier for applications where high voltage isolation is required. The differential output with a full-scale voltage of 4 V offers high immunity to noise and allows connection to a wide range of analog-to-digital converters (ADCs) powered on a 5-V nominal supply.

9.2 Typical Application

Isolated amplifiers are often used for data acquisition in industrial applications to safely separate the low-voltage portion of the system from the high common-mode voltage input of the system. The input structure of the ISO224 is optimized for isolated voltage sensing in this kind of application.

Figure 47 shows a typical operation of the device for voltage sensing as used in power line monitoring systems. The phase voltage amplitude is reduced with a resistive divider to match the input voltage range of the ISO224. The high input voltage range and the high common-mode transient immunity of the device ensure reliable and accurate operation even in high-noise environments.

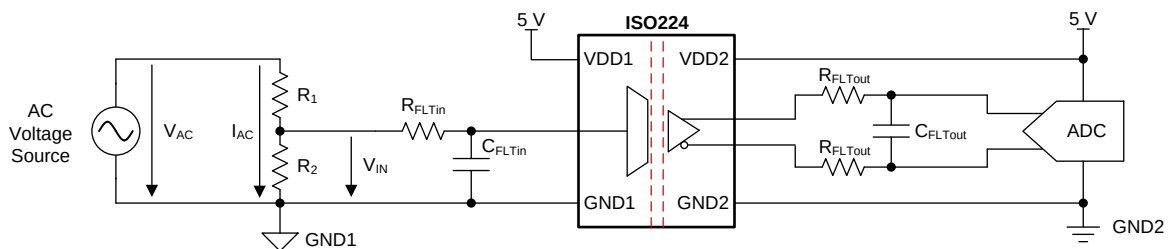


Figure 47. Using the ISO224 for AC Voltage Sensing

9.2.1 Design Requirements

Table 1 summarizes the typical design requirements for an AC power line voltage sensing application.

Table 1. Design Requirements

PARAMETER	VALUE
AC voltage range, V_{AC}	50 V to 750 V
Bandwidth	600 Hz (minimum)
Current through the resistive divider, I_{AC}	1 mA (maximum)

9.2.2 Detailed Design Procedure

The high-side power supply (VDD1) for the device is generated with a suitable isolated power source. An example of such a circuit is provided in the [Power Supply Recommendations](#) section.

The floating ground reference (GND1) is derived from one of the ends of the shunt resistor that is connected to the negative input of the device (VINN). If a four-pin shunt is used, the inputs of the device are connected to the inner leads and GND1 is connected to one of the outer shunt leads.

Use Ohm's Law to calculate the minimum total resistance of the resistive divider to limit the cross current I_{AC} to the desired value: $R_1 + R_2 = V_{AC} / I_{AC}$. The input voltage at the ISO224 results from the resistance ratio of R_1 and R_2 and the actual AC voltage: $V_{IN} = V_{AC} \times R_2 / (R_1 + R_2)$.

Consider the following two restrictions to choose the proper value of the R_1 and R_2 resistors:

- The voltage drop on R_2 caused by the nominal AC voltage range of the system must not exceed the recommended input voltage range V_{IN} of the ISO224
- The voltage drop on R_2 caused by the maximum allowed system overvoltage must not exceed the input voltage that causes a clipping output: $V_{IN} \leq V_{Clipping}$

[Table 2](#) lists examples of nominal E96-series (1% accuracy) resistor values for AC systems using 120 V, 240 V, and 400 V as nominal voltages.

Table 2. Resistor Value Examples

PARAMETER	120-V _{AC} SYSTEM	240-V _{AC} SYSTEM	400-V _{AC} SYSTEM
Resistive divider resistor R_1	115 k Ω	237 k Ω	392 k Ω
Resistive divider resistor R_2	12.7 k Ω	12.4 k Ω	12.1 k Ω
Resulting current through resistive divider I_{AC}	0.93 mA	0.93 mA	0.98 mA
Resulting input voltage V_{IN}	± 11.934 V	± 11.933 V	± 11.977 V

For systems using single-ended input ADCs with a 5-V supply, [Figure 48](#) shows an example of a TLV6001-based signal conversion and filter circuit. Tailor the bandwidth of this filter stage to the bandwidth requirement of the system and use NP0-type capacitors for best performance.

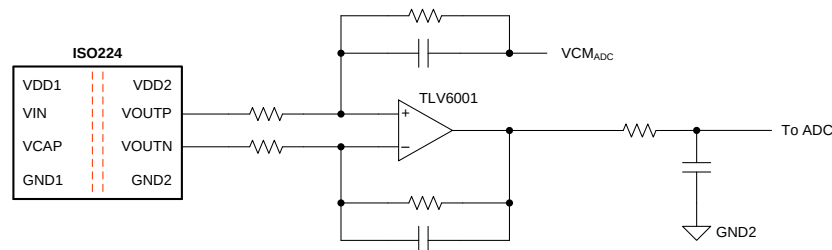


Figure 48. Connecting the ISO224 Output to a Single-Ended Input 5-V ADC

For systems using single-ended, ± 10 -V input ADCs, the [ISO224EVM](#) offers a signal path based on an [OPA277](#) that converts the differential output of the ISO224 and limits the signal bandwidth to 50 kHz.

For more information on the general procedure to design the filtering and driving stages of SAR ADCs, consult the [18-Bit, 1MSPS Data Acquisition Block \(DAQ\) Optimized for Lowest Distortion and Noise](#) and [18-Bit Data Acquisition Block \(DAQ\) Optimized for Lowest Power](#) TI Precision Designs, available for download at [www.ti.com](#).

9.2.3 Application Curves

In some applications the system must be protected in case of an overvoltage condition. To allow for fast powering off of the system, a low delay caused by the isolated amplifier is required. [Figure 49](#) shows the typical full-scale step response of the device. Consider the delay of the required window comparator and the MCU to calculate the overall response time of the system.

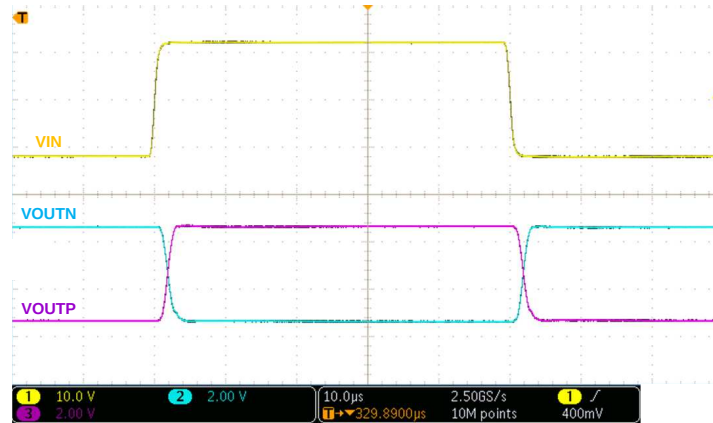


Figure 49. Step Response of the ISO224

[Figure 50](#) shows the typical AC response of the device with a full-scale sine wave with a frequency of 20 kHz applied at the input.

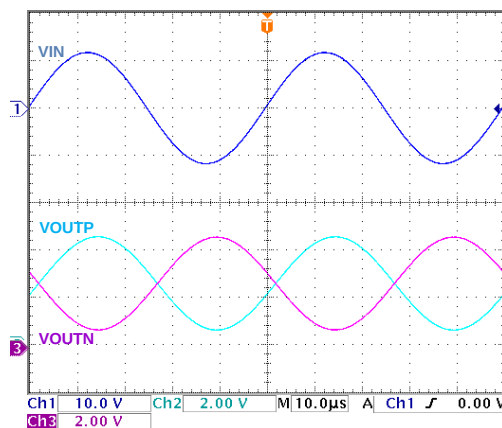


Figure 50. AC Response of the ISO224 at $f_{IN} = 20$ kHz

9.3 What to Do and What Not to Do

Do not leave the input of the ISO224 unconnected (floating) when the device is powered up. If the device input is left floating, both outputs are at the common-mode output voltage level V_{CMout} as specified in the [Switching Characteristics](#) table. See the [ISO224 Fail-Safe Output Feature application report](#) for more details.

10 Power Supply Recommendations

In a typical application, the high-side power supply (VDD1) for the ISO224 is generated from the low-side supply (VDD2) of the device by an isolated DC/DC converter circuit. A low-cost solution is based on the push-pull driver [SN6501](#) and a transformer that supports the desired isolation voltage ratings. TI recommends using a low-ESR decoupling capacitor of 0.1 μF and an additional capacitor of a minimum 1 μF for both supplies of the ISO224. [Figure 51](#) shows the recommended placement of these decoupling capacitors as close as possible to the ISO224 power-supply pins to minimize supply current loops and electromagnetic emissions.

To decouple the output of the integrated LDO, use a 0.22- μF capacitor placed as close to the VCAP pin of the ISO224 as possible.

The ISO224 does not require any specific power up sequencing. Consider the analog settling time t_{AS} as specified in the [Switching Characteristics](#) table after ramp up of the VDD1 high-side supply.

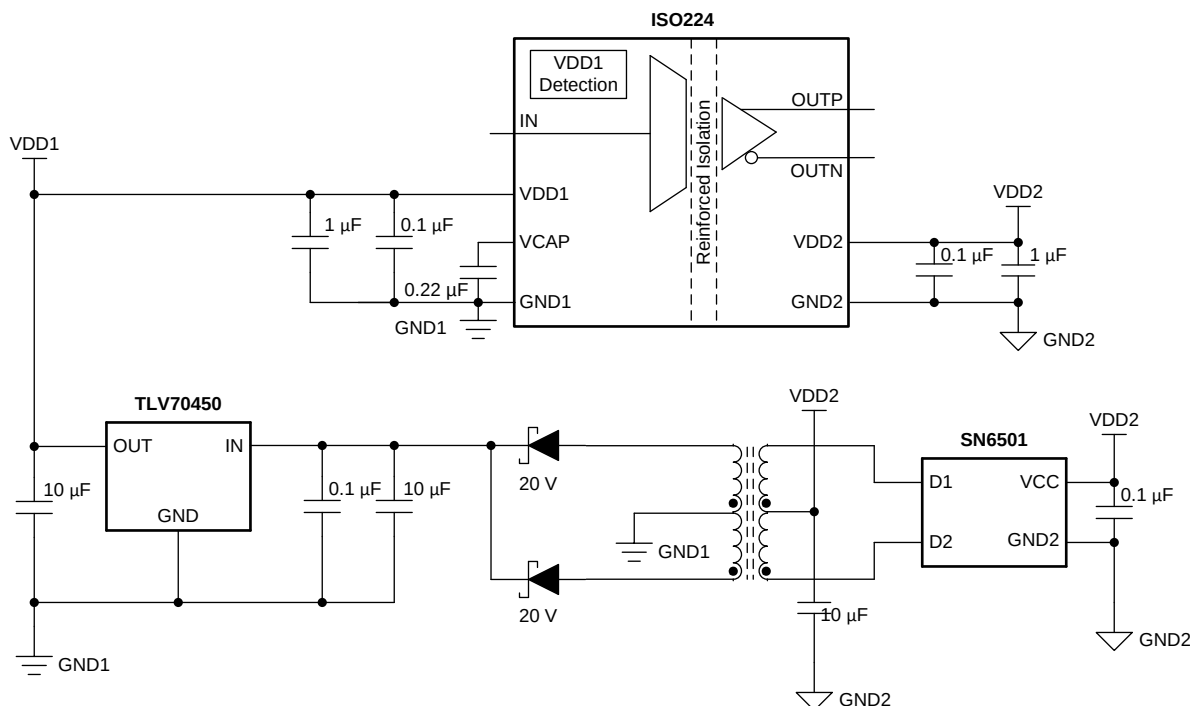


Figure 51. SN6501-Based, High-Side Power Supply

11 Layout

11.1 Layout Guidelines

For best performance, place the 0.22- μ F capacitor (C7, as shown in [Figure 52](#)) required for decoupling of the internal LDO output as close as possible to the ISO224 VCAP pin. The 0.1- μ F ceramic decoupling capacitors for both power supplies (C8 and C9) are located as close as possible to the corresponding VDDx pins followed by the additional 1- μ F ceramic capacitors for lower-frequency decoupling (C3 and C12). The resistor and capacitor used for the analog input (R1 and C2) are placed next to the decoupling capacitors. For best performance, use 0603-size or 1206-size, SMD-type, ceramic capacitors with low ESR. Connect the supply voltage sources in a way that allows the supply current to flow through the pads of the decoupling capacitors before powering the device.

[Figure 52](#) shows this approach as implemented on the [ISO224EVM](#). Capacitors C3 and C8 decouple the high-side supply VDD1 and capacitors C9 and C12 are used to support the low-side supply VDD2 of the ISO224.

11.2 Layout Example

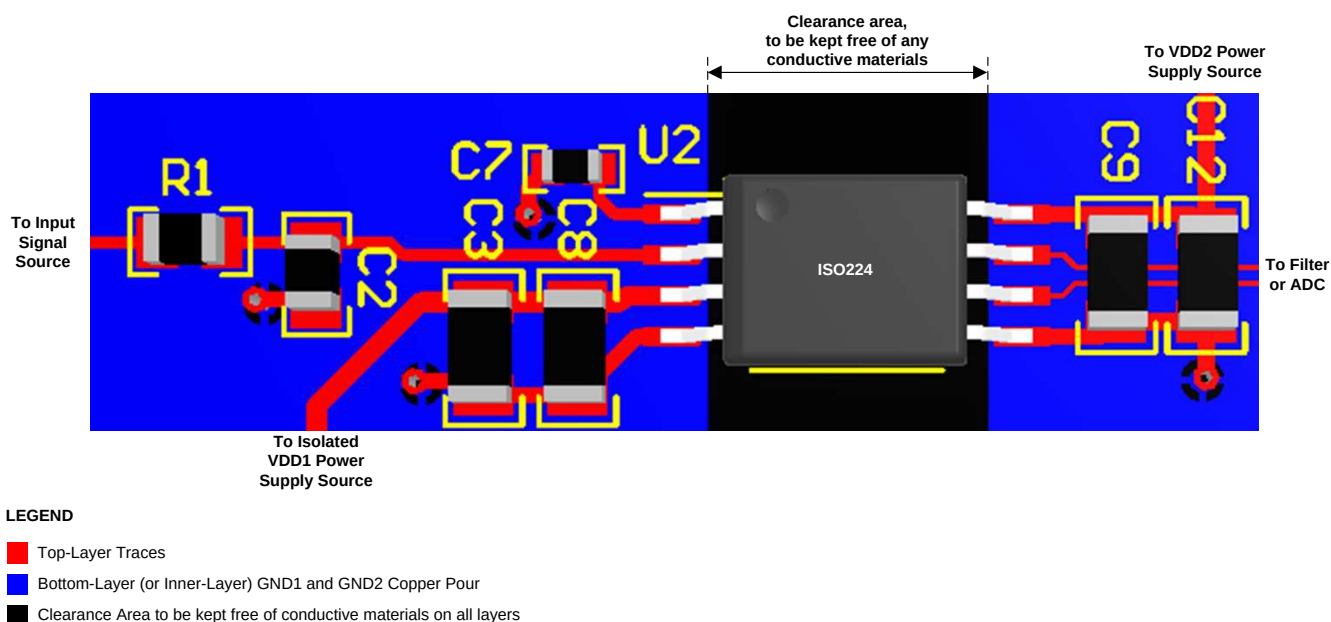


Figure 52. Recommended Layout of the ISO224

12 Device and Documentation Support

12.1 Documentation Support

12.1.1 Related Documentation

For related documentation, see the following:

- Texas Instruments, [Isolation Glossary application report](#)
- Texas Instruments, [ADS794x 14-Bit, 2 MSPS, Dual-Channel, Differential/Single-Ended, Ultralow-Power Analog-to-Digital Converters data sheet](#)
- Texas Instruments, [Semiconductor and IC Package Thermal Metrics application report](#)
- Texas Instruments, [ISO72x Digital Isolator Magnetic-Field Immunity application report](#)
- Texas Instruments, [ISO224 Fail-Safe Output Feature application report](#)
- Texas Instruments, [TLV600x Low-Power, Rail-to-Rail In/Out, 1-MHz Operational Amplifier for Cost-Sensitive Systems data sheet](#)
- Texas Instruments, [OPAx277 High Precision Operational Amplifiers data sheet](#)
- Texas Instruments, [18-Bit, 1MSPS Data Acquisition Block \(DAQ\) Optimized for Lowest Distortion and Noise TI design](#)
- Texas Instruments, [18-Bit Data Acquisition Block \(DAQ\) Optimized for Lowest Power TI design](#)
- Texas Instruments, [SN6501 Transformer Driver for Isolated Power Supplies data sheet](#)
- Texas Instruments, [TLV704 24-V Input Voltage, 150-mA, Ultralow I_Q Low-Dropout Regulators data sheet](#)
- Texas Instruments, [ISO224EVM Evaluation Module users guide](#)

12.2 Receiving Notification of Documentation Updates

To receive notification of documentation updates, navigate to the device product folder on ti.com. In the upper right corner, click on *Alert me* to register and receive a weekly digest of any product information that has changed. For change details, review the revision history included in any revised document.

12.3 Community Resources

The following links connect to TI community resources. Linked contents are provided "AS IS" by the respective contributors. They do not constitute TI specifications and do not necessarily reflect TI's views; see TI's [Terms of Use](#).

TI E2E™ Online Community *TI's Engineer-to-Engineer (E2E) Community*. Created to foster collaboration among engineers. At e2e.ti.com, you can ask questions, share knowledge, explore ideas and help solve problems with fellow engineers.

Design Support *TI's Design Support* Quickly find helpful E2E forums along with design support tools and contact information for technical support.

12.4 Trademarks

E2E is a trademark of Texas Instruments.
All other trademarks are the property of their respective owners.

12.5 Electrostatic Discharge Caution



These devices have limited built-in ESD protection. The leads should be shorted together or the device placed in conductive foam during storage or handling to prevent electrostatic damage to the MOS gates.

12.6 Glossary

SLYZ022 — *TI Glossary*.

This glossary lists and explains terms, acronyms, and definitions.

13 Mechanical, Packaging, and Orderable Information

The following pages include mechanical, packaging, and orderable information. This information is the most current data available for the designated devices. This data is subject to change without notice and revision of this document. For browser-based versions of this data sheet, refer to the left-hand navigation.

PACKAGING INFORMATION

Orderable part number	Status (1)	Material type (2)	Package Pins	Package qty Carrier	RoHS (3)	Lead finish/ Ball material (4)	MSL rating/ Peak reflow (5)	Op temp (°C)	Part marking (6)
ISO224ADWV	Active	Production	SOIC (DWV) 8	64 TUBE	Yes	NIPDAU	Level-3-260C-168 HR	-55 to 125	ISO224A
ISO224ADWVR	Active	Production	SOIC (DWV) 8	1000 LARGE T&R	Yes	NIPDAU	Level-3-260C-168 HR	-55 to 125	ISO224A
ISO224ADWVRG4	Active	Production	SOIC (DWV) 8	1000 LARGE T&R	Yes	NIPDAU	Level-3-260C-168 HR	-55 to 125	ISO224A
ISO224BDWV	Active	Production	SOIC (DWV) 8	64 TUBE	Yes	NIPDAU	Level-3-260C-168 HR	-55 to 125	ISO224B
ISO224BDWVR	Active	Production	SOIC (DWV) 8	1000 LARGE T&R	Yes	NIPDAU	Level-3-260C-168 HR	-55 to 125	ISO224B
ISO224BDWVRG4	Active	Production	SOIC (DWV) 8	1000 LARGE T&R	Yes	NIPDAU	Level-3-260C-168 HR	-55 to 125	ISO224B

⁽¹⁾ **Status:** For more details on status, see our [product life cycle](#).

⁽²⁾ **Material type:** When designated, preproduction parts are prototypes/experimental devices, and are not yet approved or released for full production. Testing and final process, including without limitation quality assurance, reliability performance testing, and/or process qualification, may not yet be complete, and this item is subject to further changes or possible discontinuation. If available for ordering, purchases will be subject to an additional waiver at checkout, and are intended for early internal evaluation purposes only. These items are sold without warranties of any kind.

⁽³⁾ **RoHS values:** Yes, No, RoHS Exempt. See the [TI RoHS Statement](#) for additional information and value definition.

⁽⁴⁾ **Lead finish/Ball material:** Parts may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead finish/Ball material values may wrap to two lines if the finish value exceeds the maximum column width.

⁽⁵⁾ **MSL rating/Peak reflow:** The moisture sensitivity level ratings and peak solder (reflow) temperatures. In the event that a part has multiple moisture sensitivity ratings, only the lowest level per JEDEC standards is shown. Refer to the shipping label for the actual reflow temperature that will be used to mount the part to the printed circuit board.

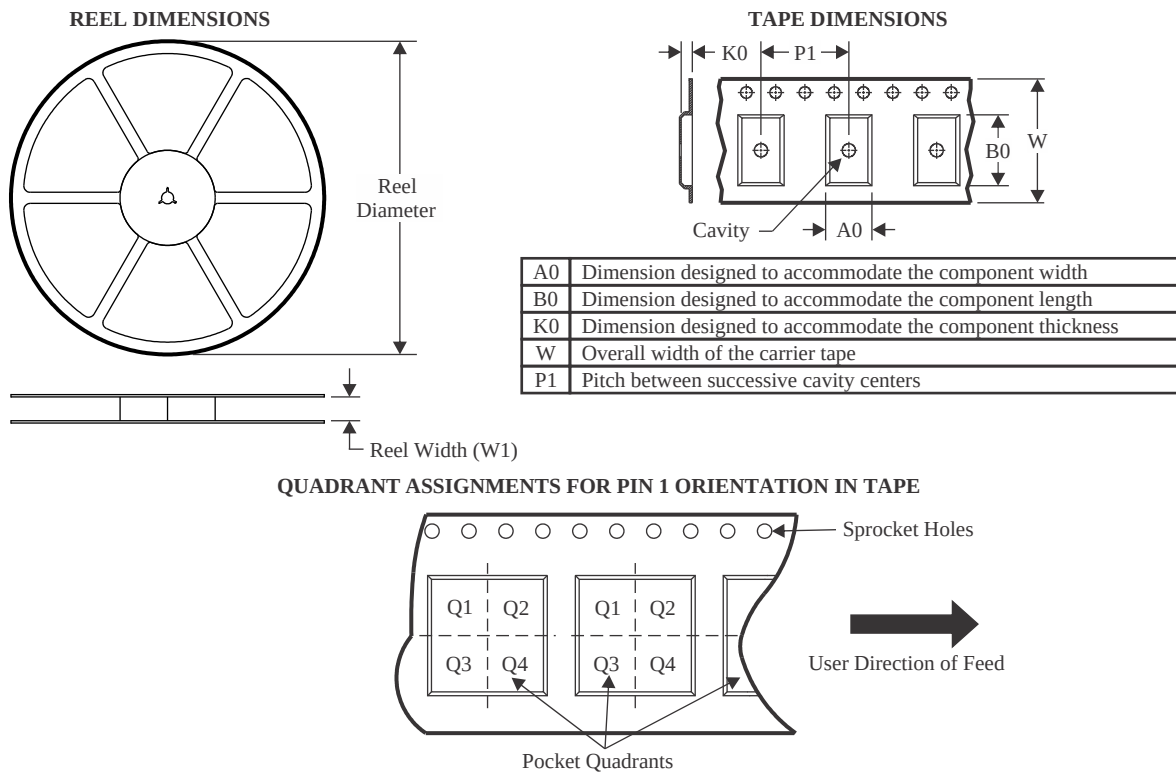
⁽⁶⁾ **Part marking:** There may be an additional marking, which relates to the logo, the lot trace code information, or the environmental category of the part.

Multiple part markings will be inside parentheses. Only one part marking contained in parentheses and separated by a "~" will appear on a part. If a line is indented then it is a continuation of the previous line and the two combined represent the entire part marking for that device.

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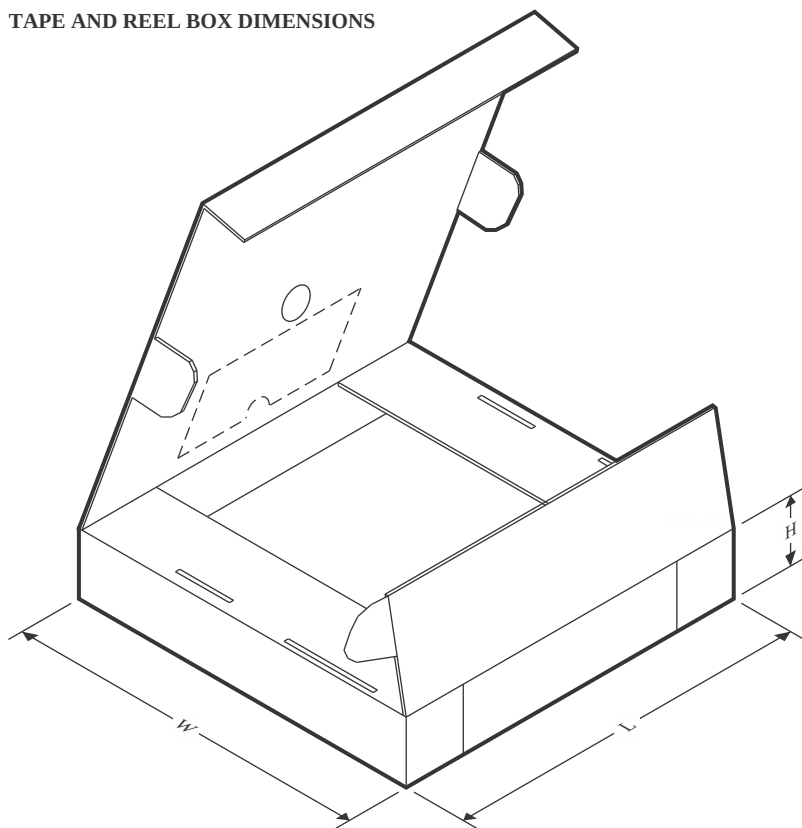
TAPE AND REEL INFORMATION



*All dimensions are nominal

Device	Package Type	Package Drawing	Pins	SPQ	Reel Diameter (mm)	Reel Width W1 (mm)	A0 (mm)	B0 (mm)	K0 (mm)	P1 (mm)	W (mm)	Pin1 Quadrant
ISO224ADWVR	SOIC	DWV	8	1000	330.0	16.4	12.05	6.15	3.3	16.0	16.0	Q1
ISO224ADWVRG4	SOIC	DWV	8	1000	330.0	16.4	12.05	6.15	3.3	16.0	16.0	Q1
ISO224BDWVR	SOIC	DWV	8	1000	330.0	16.4	12.05	6.15	3.3	16.0	16.0	Q1
ISO224BDWVRG4	SOIC	DWV	8	1000	330.0	16.4	12.05	6.15	3.3	16.0	16.0	Q1

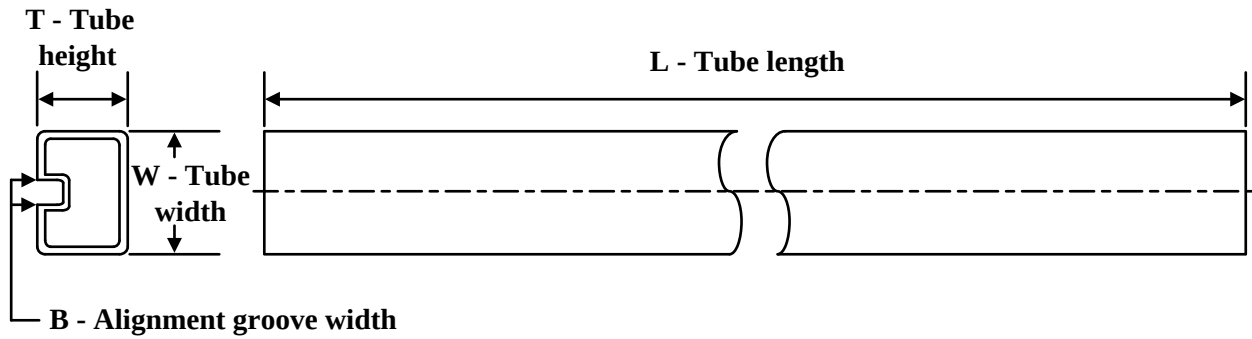
TAPE AND REEL BOX DIMENSIONS



*All dimensions are nominal

Device	Package Type	Package Drawing	Pins	SPQ	Length (mm)	Width (mm)	Height (mm)
ISO224ADWVR	SOIC	DWV	8	1000	350.0	350.0	43.0
ISO224ADWVRG4	SOIC	DWV	8	1000	350.0	350.0	43.0
ISO224BDWVR	SOIC	DWV	8	1000	350.0	350.0	43.0
ISO224BDWVRG4	SOIC	DWV	8	1000	350.0	350.0	43.0

TUBE

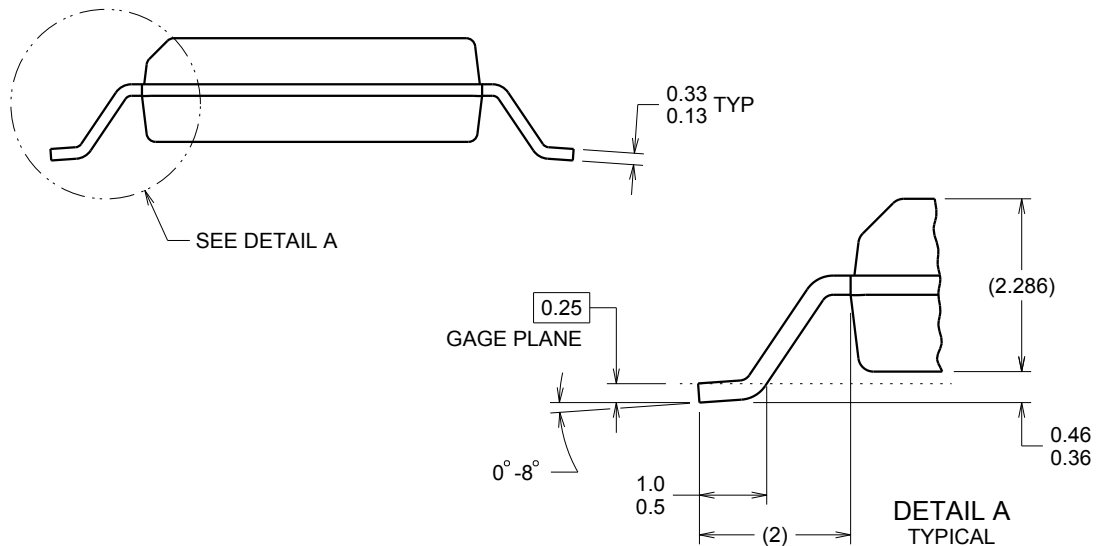


*All dimensions are nominal

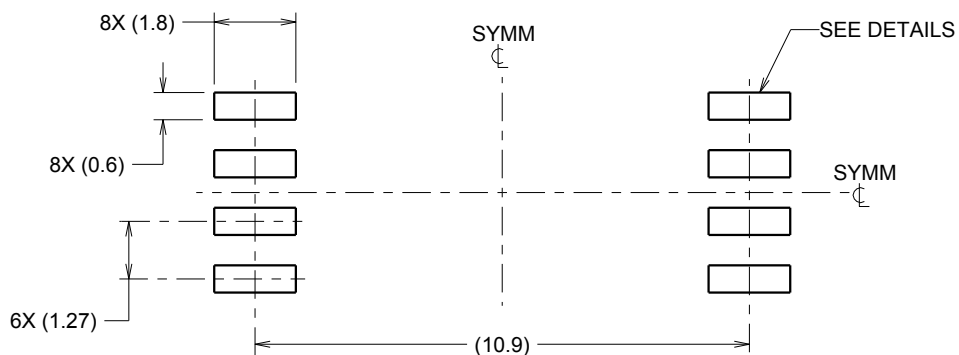
Device	Package Name	Package Type	Pins	SPQ	L (mm)	W (mm)	T (μm)	B (mm)
ISO224ADWV	DWV	SOIC	8	64	505.46	13.94	4826	6.6
ISO224BDWV	DWV	SOIC	8	64	505.46	13.94	4826	6.6

SOIC - 2.8 mm max height

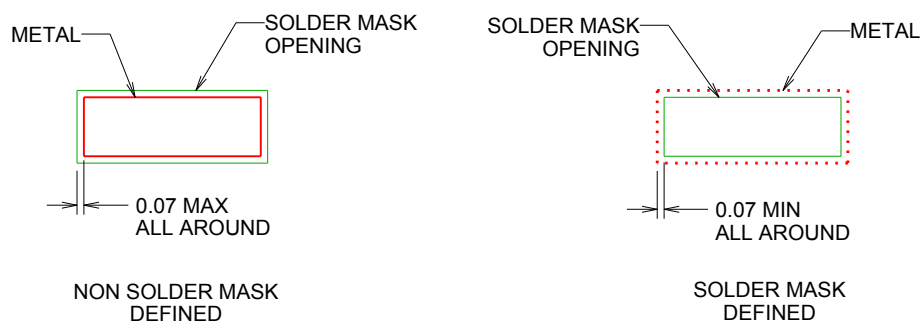
PIN 1 ID AREA
 11.5 ± 0.25 TYP
 5.95
 5.75
 NOTE 3
 1
 4
 8
 5
 6X 1.27
 2X 3.81
 8X 0.51
 0.31
 7.6
 7.4
 NOTE 4
 A
 B
 SEATING PLANE
 0.1 C
 2.8 MAX
 C
 A
 B
 S



1. All linear dimensions are in millimeters. Dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.15 mm, per side.
4. This dimension does not include interlead flash. Interlead flash shall not exceed 0.25 mm, per side.



LAND PATTERN EXAMPLE
9.1 mm NOMINAL CLEARANCE/CREEPAGE
SCALE:6X

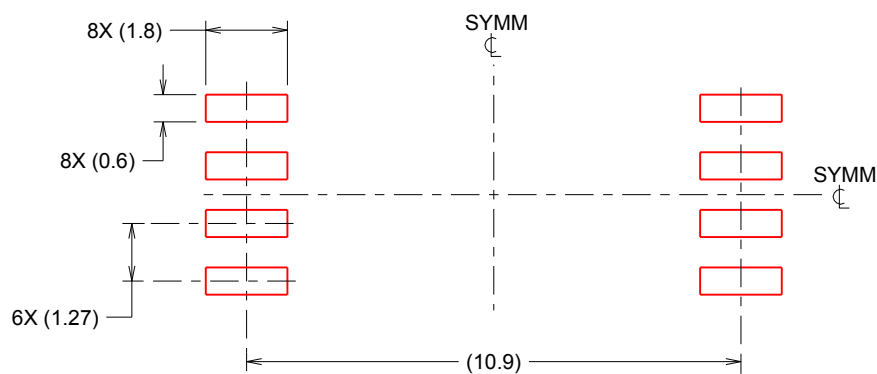


SOLDER MASK DETAILS

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NOTES: (continued)

5. Publication IPC-7351 may have alternate designs.
6. Solder mask tolerances between and around signal pads can vary based on board fabrication site.



SOLDER PASTE EXAMPLE
 BASED ON 0.125 mm THICK STENCIL
 SCALE:6X

4218796/A 09/2013

NOTES: (continued)

7. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
8. Board assembly site may have different recommendations for stencil design.

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